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HERALD COLLEGE KATHMANDU

6CS007- Project and Professionalism

RouteMate (A Bus Route Tracking System)

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Submitted on : 19th May, 2025

Word Count : 8886

Project Title

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Award Title: ***BSc(Hons) Computer Science***

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Abstract

The Bus Route Tracking System displays the actual bus locations in real-time using real time location. The system tracks buses by processing mobile network signals or Wi-Fi. The system operates through two sections: Node.js develops the backend data processing while the frontend design with React.js shows bus updates to users.

The project targets these locations because it wants to improve public transport reliability and ease of access especially when GPS is out of reach or cost-prohibitive. Gantt charts was used to check the progress toward deadlines. Real-time bus tracking functions through basic available information sources to demonstrate that sophisticated technology is not a prerequisite for this capability.

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1. Introduction

1.1 Project briefing

In cities like Kathmandu, managing and tracking public transportation is a big challenge. Public transport often has problems like irregular schedules, poor communication and no real time updates. These issues make people lose trust in public transport and use private vehicles instead, causing more traffic and pollution.

The Bus Route Tracking System, called RouteMate, is designed to make public transportation easier to use with advanced mobile technology. It provides real-time updates, route suggestions, and a simple interface, helping passengers plan their trips easily and confidently. RouteMate improves communication between passengers and transportation services by giving accurate and timely information.

Key Features:

- Bus tracking using real time data.
- Route navigation based on user location and preferences.
- Locates the nearest bus stop.
- Digital ticketing.
- Contactless payment.
- Notifies emergency services in case of accident.

By addressing common issues such as not knowing bus location, difficulty in locating the nearest bus stop, route suggestions for new residents and uncertainty about arrivals times, RouteMate help users enjoy a smoother and more efficient travel experience. Furthermore, the app caters to diverse user groups, ensuring accessibility and ease of use for commuters of all age groups and technical proficiencies.

1.2 Aims

- Create a user-friendly system where passengers can easily find and track bus routes in real time.
- Include map integration to help users locate bus stops and routes near them.
- Ensure secure user data handling and a smooth experience for both passengers and operators.
- Provide estimated arrival and departure time to help passengers plan better.
- Allow passengers to save favorite routes or frequent used stops for quick access.
- Allow users to give a feedback and rate their travel experience.
- Enable passengers to cancel route navigation or bookings within a certain timeframe if needed.

1.3 Objectives

- Learning about the React native framework for mobile applications.
- Understand how PostgreSQL works and how to use it for managing database
- Gain a good understanding of JavaScript and its related tools.
- Create an easy-to-use and efficient bus route tracking system for smooth navigation.
- Ensure the bus route tracking system is fully responsive across all devices.
- Solving the problem of public transport location tracking.
- Implementing an online bus ticketing platform.
- Improve public transport accessibility.

1.4 Artefact

1.4.1 Functional Decomposition Diagram

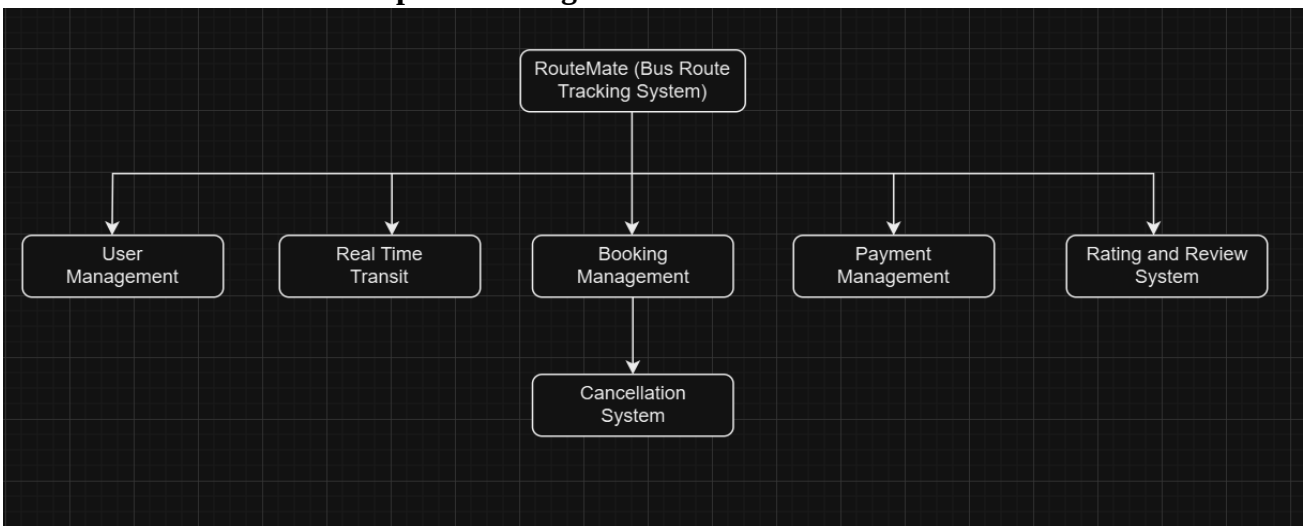


Figure 1: Functional Decomposition Diagram

RouteMate is meant to enhance your experience with bus travel. User Management manages account and profile information and Real-Time Transit is there to help you track when the bus will arrive. You can easily manage your reservations and any changes, thanks to the Cancellation System. The system for payments is safe and the Rating and Review System allows riders to describe their journeys. All these systems function together to ensure the bus runs more smoothly and reliably.

1.4.2 Systems and Sub System

User Management

Users in RouteMate access secure registration and login services through the User Management Subsystem which also enables role changes. Registering on RouteMate requires users to create their own username and select a secure password for their account together with an email. The platform provides visibility options for passwords as well as automatic password reset functions. Among the sequence of events after registration the system delivers verification and follows with an automatic move to login.

Users can enable Driver Mode to accomplish document validation and pick a vehicle from a list. The authentication system coupled with role transition functions combine to provide efficient user-friendly operations throughout RouteMate platform.

Real-time Transit System

The Real-Time Transit System (RT-TS) serves users and administrators by showing current bus positions as well as arrival and departure times in real-time. The system depends on driver activation of their location sharing so the real-time data becomes available for use in accurate bus tracking through interactive maps. The system displays estimated arrival times (ETAs) according to dynamic traffic information for users who enact route selection by stop or number.

The system organizes bus displays for users based on their preferences so they receive personalized Trip information. Real-time tracking enables smooth trip planning since it is demonstrated in the "Track Bus in Real-Time" use case which enables passengers to track buses' movements and forecast their arrival times without interruption.

Booking Management System

Users of the Booking Management System have the capability to reserve bus tickets through an interface which allows them to pick their routes and seats and travel times. Users receive error alerts when they forget essential information and bookings lead to confirmation messages containing trip specifics. User accounts enable browsing and termination of bookings (between 20 minutes before departure yet exclude reimbursement opportunities) in addition to showing booking records. Admin users have access to all reservations while drivers can only see the lists of passengers on each of their routes. The refund process might extend to multiple days before the payment can be returned to users. Users receive notifications for successful payment transfers that additionally show their travel information.

Payment Management

Users can engage in safe transactions through Payment Management System whereas eSewa and Khalti and other payment features join forces with secure encrypted payment protocols. Users pay with contactless methods that include QR codes and instantly receive payment status notifications. Users along with admins maintain access to payment information and users are able to monitor ticket payment statuses after booking. Users achieve secure payment transactions through a system that provides safe and convenient as well as transparent payment methods

Rating and Review Management

This system allows customers to share how their bus experiences were. Ratings from users help others higher their ride-sharing experience. It also helps the company improve the service it offers customers. Trust is built and all participants know the situation is fair.

1.5 Academic Question

How can a mobile-based solution improve the efficiency and reliability of public transportation routes in urban areas?

Mobility solutions through an app enhance city bus quality by providing accurate instant information to commuters. People utilizing GPS tracking receive exact bus location data along with updated arrival times thus avoiding longer than necessary waiting times. The application enables bus companies to find alternative routes based on traffic conditions or delays which improves operational efficiency of the entire system.

The system enables users to report their opinions by informing the network about overcrowding issues or late bus arrivals. Transportation managers receive immediate help to fix problems through this system. Using digital tickets together with trip planning features enables bus passengers to hike convenience and speed

up their travel experience. Public transport will gain increased usage among passengers when buses operate efficiently and maintain easy access to services thus reducing city traffic and its related pollution. Public transport improves under the combined benefits of a bus-tracking app which benefits both users and operators thereby enhancing operational reliability.

1.6 Scope and limitation

1.6.1 Scope

RouteMate Solution helps public transport with live tracking, better routes, and easy digital payments. Passengers and transportation operators obtain beneficial improvements from this system through accurate tracking data along with prompt arrival updates and best travel directions. The system improves user experience through its functionality that lets commuters see bus stops near them and develop better travel plans. Digital ticketing combined with contactless payments brings efficiency to transactions so clients no longer need to rely on cash and enjoy more straightforward service. Calling for emergencies on the platform enables RouteMate to warn relevant first responders through an emergency notification system. The system that utilizes modern mobile technology enhances transport service-patron communication and advances public transport usage which results in lower traffic congestion and decreases pollution levels.

1.6.2 Limitation

- Requires internet access for real-time tracking and updates
- Accuracy of live location tracking depends on GPS and network availability
- Limited adoption if transport operators do not integrate the system
- Initial setup and maintenance costs may be high
- User acceptance and familiarity with digital ticketing may take time
- Potential privacy concerns related to location tracking
- System performance may be affected by technical issues such as server downtime

1.7 Report Structure

- **Background**

The background part gives readers a clear picture of the project. It gives details about the project's goal and the audience for whom it is developed. The writer explains the project further by pointing out how it is similar to others and highlighting what it does and how it functions.

- **Development**

The development section allows readers to grasp how the project was progressed. It ensures you are given the instructions and steps to complete the project properly at each stage of doing the method. Includes the diagrams important to the project along with an in-depth look at the surveys conducted. It also discusses in brief the key features and layout of the project.

- **Testing and Analysis**

In the testing and analysis section, each test stage of the project is explained with its evidence included. It also explains in detail the way all the elements of the system are interconnected. The purpose of this part is to uncover any defects in the system. The logic in systems is tested to confirm that all goals have been met.

- **Conclusion**

The last section of the report offers an overview of the project. It explains a few of the difficulties that arise when implementing a project. The following part covers the pros and cons of the project. In addition, it discusses what could happen in the future to improve or strengthen the project. This part shows the readers what the work actually produced. It also demonstrates where the project can progress or improve in the future.

2. Literature Review

2.1 Research Papers

i. Smart Urban Mobility: Bus Tracking Systems

Smart cities have been discussed as urban populations increase, and because there must be an effective and lasting solution to issues such as transport. Modern smart cities utilize ICT to improve the structures in place, with focus on transport by using data in real time. The paper, "Towards Smart Mobility in Urban Spaces: From the paper titled, ‘’Development of Bus Tracking and Information Application” authored by Wong Seng Yue, Koh Keng Chye, and Cheong Wan Hoy proposes the creation of a bus tracking and information system (BTA) to effectively solve inefficiencies that plague urban transportation in Malaysia. The proposed system ensures that the bus location and scheduling are precise, with the use of GPS technology, and the use of international standard such as GTFS. They can help to decrease the time spent by individuals on waiting for transport hence empowering the government by reducing anthropological influence in that particular area especially with regards to the misuse of automobiles.

The advantages of BTA systems include increased user satisfaction, administrative benefits, and satisfied environmental impacts Nevertheless, several limitations exist with the implementation of such systems. The only potential problem is the rapid dependence on bus agencies for real-time data that can delay the working deployment, and the problems like, for example, fluctuating GPS signals need to be tackled using out-of-the-box approaches like crowd-sourcing. Nevertheless, applications of BTA are a way forward towards attaining sustainable development of cities. In this context, they are consistent with global tendencies in smart mobility by enhancing public transportation and contributing to the achievement of smart city goals. Future work should extend to accumulating more intricate algorithms to increase the precision of data and incorporating end-users’ input to improve the system’s usability and operation. (Yue, Chye, & Hoy, 2017)

ii. The Role and implementation of Digital Payment system in Nepal

This research paper looks at how digital payment systems are making it easier for Nepalese people and also how apps like eSewa Khalti and Fonepay are replacing old methods like cash handovers or back transfers with faster cheaper and more convenient ways to send money. These systems are especially important for people in rural areas who don't have easy access to banks. They also bring other benefits, like helping families save money and plan better for the future. But not everything is perfect, problems like limited internet access, low trust in digital platforms and lack of tech knowledge make it harder for everyone to use these services.

To add payment options like Esewa or Khalti to others app, developers can connect the app these services using their APIs. This allows users to pay directly within the app by simply logging in, selecting the payment method, and completing the transaction securely. With safety features like two-factor authentication, users' payments are kept secure. After the payment, the app gets confirmation and updates the transaction, making the whole process easy and safe for users. (Koirala, 2024)

iii. Learning Patterns of Activity Using Real-Time Tracking

In this work by Chris Stauffer and W. Eric L. Grimson, the reader is presented to an intelligent video tracking framework that is capable of capturing and analyzing activity patterns in a real environment. Employing the adaptive background subtraction, only moving objects are subtended from static background and recognized. The weakness of the previous techniques is that they fail in cases of variation in lighting and swaying trees whereas this system utilizes the combination of Gaussian models and is therefore efficient in efficiency in dynamic outdoor and indoor applications. The system is able to track multiple objects in motion, shadow and scene transition making it a practical solution for domestic use.

As soon as the objects are tracked, the system then identifies activity patterns and then categorizes them hierarchically. It develops a set of `attributes' like size, speed and direction of movement, to categorize activities as pedestrian or vehicular movements and so on. The system identifies odd motion, for instance when a car behaves in an unusual manner, and has been tried on roads, offices and in the wild. Although it proves adequate, it loses efficiency in scenes characterized by overlapping objects or frequent switch on directions. They also mentioned that possible enhancements of the system, which would include depth perception or contextual data will be even more accurate and useful. This research presents perspective that new video can optimize subsector such as traffic, security and behaviors. (Grimson, 2018)

iv. Improving User Experience in Bus Ticketing Booking: RedBus

This research aims at determining to what extent the effectiveness of the RedBus app can be boosted by an ideal UX. The RedBus app is the most popular for booking bus tickets in many countries, but when testing for usability, it was possible to detect some flaws in designing: confusing layouts and non-constant interface. These problems affect users in the following ways: This leads to dissatisfaction of the innovative app and thus reducing the user's motivation of continued use of the application. To address this, the research employed the

Design Thinking method, which involves five stages: This is Empathize, Define, Ideate, Prototype and test. With this structure in place, it is guaranteed that the redesign aspect will be a user-driven one and that the solutions it posits are viable as well as revolutionary. The effectiveness, efficiency and satisfaction measurement process involves usability testing in this framework. Existing researches proved this method efficiency to enhance usability of digital products.

Taking the approach into practice in the study, extensive enhancements were observed once the new app design was done. Overall; any effectiveness by 44%; efficiency by 0.079 goals per second; and user satisfaction by 63%; appreciating the implementation of user centered design. These findings are consistent with prior studies as Nielsen (2024) emphasizing the effectiveness of considering multiple cycles of the design process. This research is peculiar because of including usability testing into the Design Thinking framework; checking the guide for developers interested in improving UX. The future work of similar nature to this research could be applied to other platforms, for instance iOS or Web applications to guarantee the obtained results' universality as well as the user's long-term interest and activity. (Valencia & Lisana, 2024)

v. Perception on QR Payment in Kathmandu Valley

This research paper provides an in-depth look at how users in Kathmandu are adapting to QR payment systems. Using the UTAUT model, it identifies key factors such as performance expectations, effort expectancy, social influence and facilitating conditions that shape customer perception and adoption. Performance expectation and facilitating conditions were found to have the strongest influence, showing that customers appreciate the speed, convenience and flexibility offer by QR systems like eSewa, Fonepay and Khalti. The study also highlights that QR systems are increasingly used for tasks like fun transfer, utility bill payments and shopping, with 54.29% of respondents having used these systems for 1-3 years. The COVID-19 Pandemic accelerated their adoption as people prioritized contactless payment options. However, challenges like technical glitches, poor internet connectivity and insufficient user education remain significant barriers, affecting the smooth experience many users expect.

To overcome these challenges, the study suggests measure like regular system maintenance, stronger internet infrastructure, and targeted user training. It emphasizes the importance of Collaboration with payment processors and secure platforms to build trust and reliability among users. The study concludes that with improvements in technical reliability and customer support, QR payments have the potential to become an integral part of Nepal's financial ecosystem, making transaction more seamless and accessible. The research offers valuable insights for policymakers, developers and service providers to further enhance digital payment adoption and user satisfaction in the Kathmandu valley. (Alina Maharjan, 2024)

vi. Research on Real-Time Transit Systems

Real-time transit systems play a crucial role in closing the communication gap between transport operators and passengers. Studies conducted by Ferris et al. (2010) highlight that tools like OneBusAway significantly enhance commuter satisfaction by minimizing the uncertainty associated with waiting times. By combining GPS technology with General Transit Feed Specification, OneBusAway delivers accurate predictions for buss arrivals, which has been shown to increase passenger's confidence in public transportation services.

Research highlights the significant impact of real-time transit systems on urban mobility, with studies like Watkins et al. (2011) demonstrating how real-time tracking enhances passengers and encourages greater reliance on public transport over private vehicles. This shift not only reduces traffic congestion but also contributes to environmental sustainability. This system exemplify the potential of innovative technologies to make public transit more reliable and user-friendly. However, challenges remain in improving scalability, ensuring data accuracy and overcoming connectivity issues which present opportunities for further advancements in the field. (Gate, 2022)

vii. Research on Ride Sharing platforms

Ride-sharing applications have revolutionized transportation by utilizing technology to efficiently link drivers with passengers. According to research by Rayle et al. (2016), these platforms improve mobility, particularly in regions where traditional public transit options are limited. They offer affordable and adaptable solutions, reducing the reliance on private vehicles and making transportation more accessible to diverse users. Furthermore, these platforms provide an alternative for people without access to personal vehicles, helping bridge gaps in transportation infrastructure and promoting convenience in daily commutes.

Additionally, studies highlight the environmental advantages of ride-sharing. Carpooling features, such as those offered by Uber Pool and Lyft Line, help lower vehicle emission and alleviate traffic congestion. Clewlow and Mishra (2017) emphasize that these services enhance urban transportation efficiency by combining trips and increasing vehicle occupancy rates. Moreover, the shared model reduces the number of single-occupancy vehicles on the road, supporting sustainability goals. With advancements in technology, these platforms are expected to further integrate electric vehicles and renewable energy solutions, amplifying their environmental impact. (eScholarship, 2022)

2.2 Similar Application

i. Google Maps

Google Maps is an application, which is commonly utilized for the navigation purposes. Your maps will roadmap for driving, walking, cycling, and also travel by other means such as using public transport. The app provides directions with real-time traffic conditions and direction by turn options for users. This can also enable it to re-route in order to avoid traffic hold-up. Street view is also present in Google Maps whereby instead of pictures of interiors you can view 360 pictures of streets Technology also offers offline maps that work even when a user has no internet connection. It contains information about local businesses and users' rate, so using the app people can find interesting places. Further, it connects with other applications such as the ride hailing services to among others more travel options to make it useful in both direction and discovery. (TechTarget, 2024)

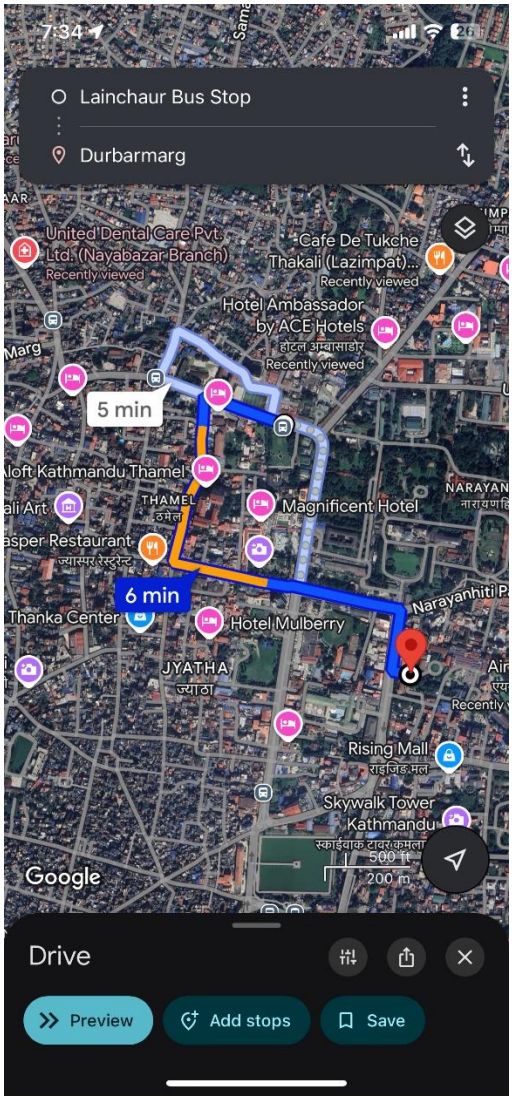


Figure 2: Google Maps

ii. Waze

Waze is an application for navigation that employs data provided by its users. They can inform their experience like traffic congestion, an auto crash, or a shutdown of any given road and this is immediate to the others. It makes users be able to skip some roads and get to their destination faster. Waze also has features such as voice commands, fuel prices and traffic divination and making changes to the directions. It has also the shared transportation, implying if one is traveling in the same direction as another, it opportunity share a vehicle and hence saving on cost and improving traffic. Since Waze is dependent on its users and is being updated in real time, it is a very helpful app to those who are driving. (Wikipedia, 2024)

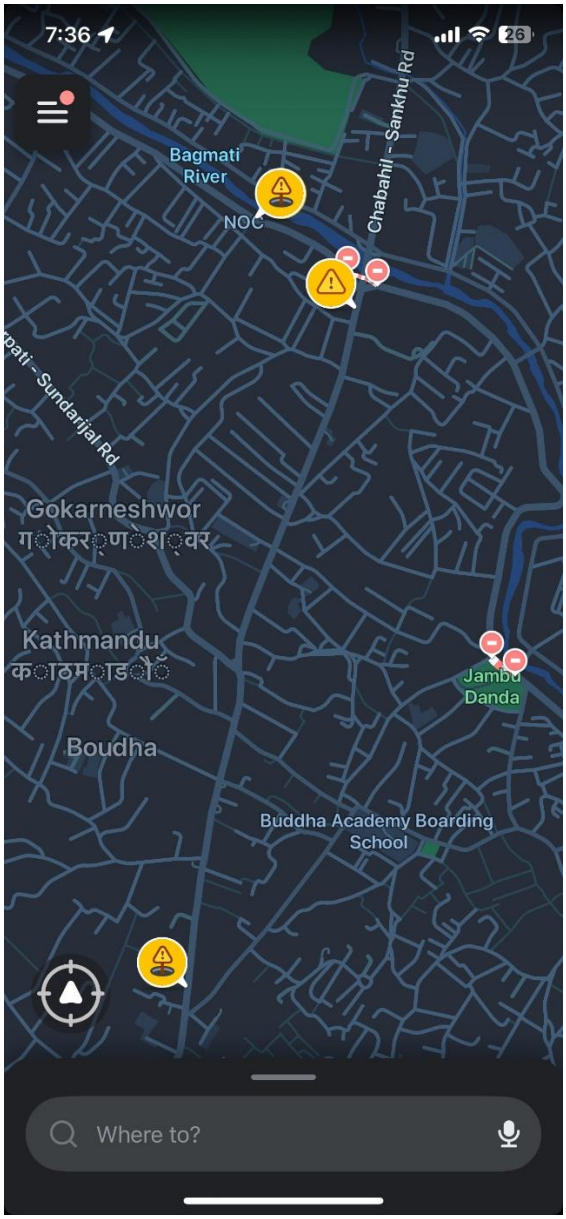


Figure 3: Waze

iii. Moovit

It is an app for public transport. Users can search for buses and trains and other means of transport in more than 300 cities around the globe. The ultimate availability of an app that can bring together the scheduler of combined travel, such as, walking, cycling and public transport into one route isle. Moovit allows users to access updates on schedules, delays and changes in real-time hence making it easy to plan. It also useful for disabled people as well as providing step-free routes and voice directions. Moovit works with other service providers including ride-hail providers such as Uber on the last segment of a journey. Its availability and design attributes make it suitable for use in public transport systems. (Moovit, 2024)

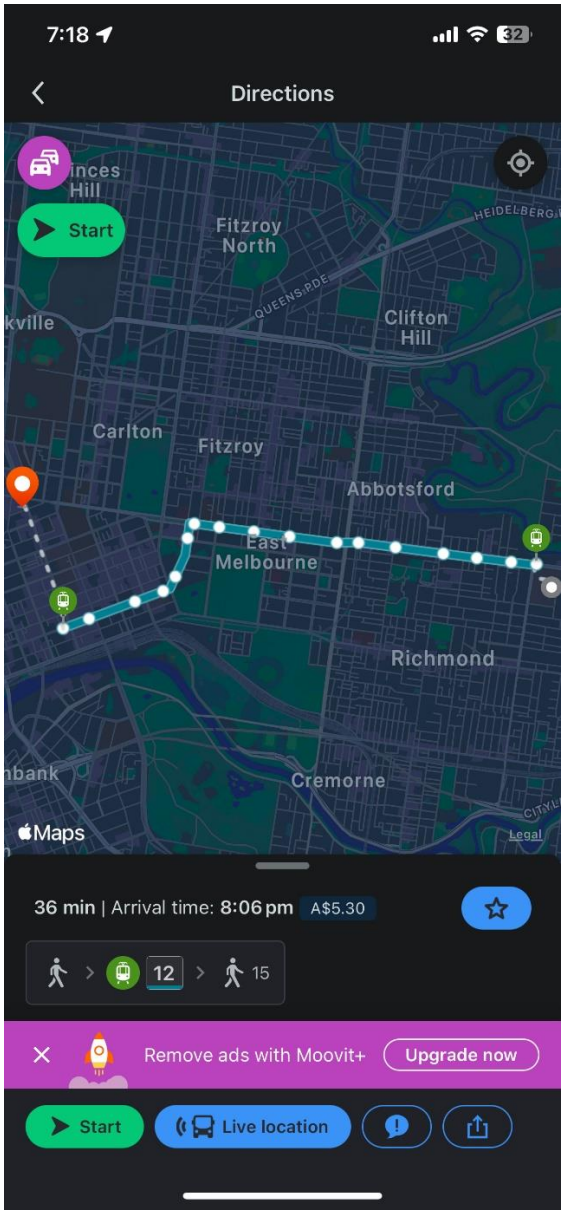


Figure 4: Moovit

iv. Uber

It is an application that assists people in getting an automobile service where they want it. You can always order a car through the app, which establishes the connection with the closest driver. It reveals to the driver where they are and how long it will take for them to get to a certain destination and so forth making the trip convenient. In this application, fares are pre-determinately estimated, therefore, a user knows the sum he/ she will pay after beginning a ride. It has a variety of services: UberX for cheap prices and Uber Black for high prices. Both drivers and passengers can rate each other, this eliminates the possibility of a driver gets paid for bad service. Uber is an option that so helpful to get around in some cities where there is no enough system of public transportation. (Mobisoft, 2024)

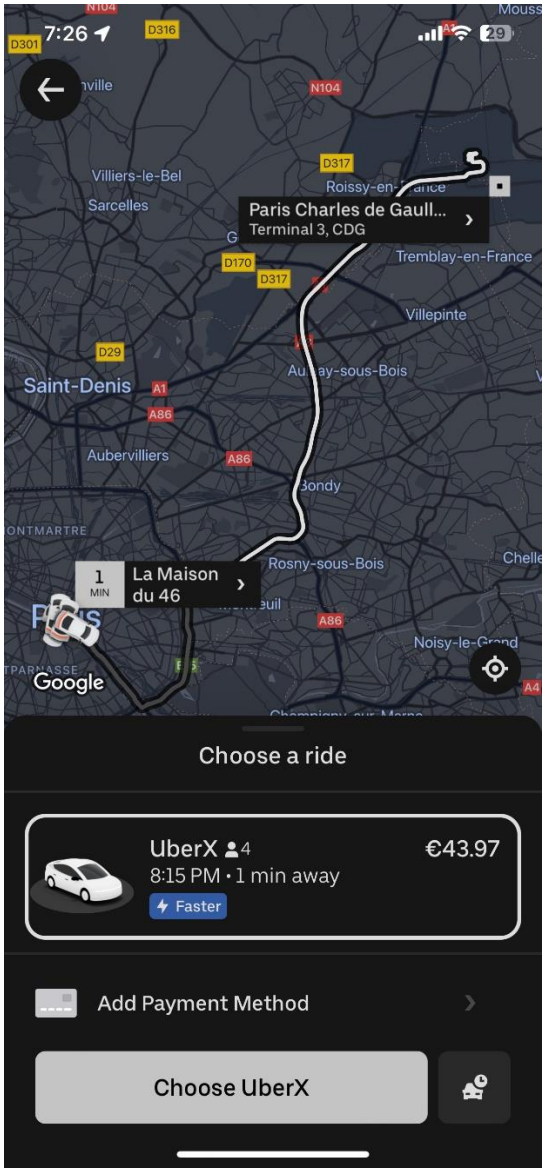


Figure 5: Uber

v. SG Buses

One of the most used applications in Singapore named ‘SG Buses’ deals with public bus lines and it provides precise real time information on arrival to the certain stops, so it is useful for organizing the trips. The app includes all the public buses that are being used by SBS Transit, SMRT and Tower Transit allowing users to search for bus services by number or via bus stop. The App features bus stop maps and also maps of locating other areas/ landmarks will make it easier for the users while in unknown territories. You can list your favorite bus stops for future reference and the simple and clean interface and layout are good for new and old users. SG Buses is useful for anyone who needs bus to commute around the Singapore because it provides accurate information.

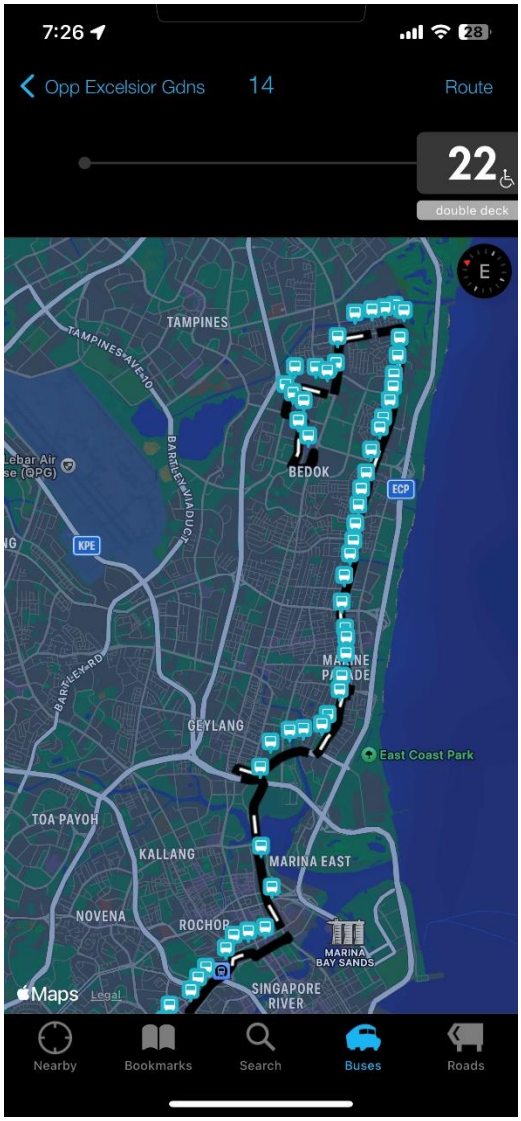


Figure 6: SG Buses

3. Project Methodology

3.1 Agile Methodology

The project selects Agile methodology to match the system requirements that continuously transform and the requirement for immediate user feedback. The proposed system needs to adjust to changing user needs and scope requirements so Agile methodology provides flexibility through its iterative development method. Throughout small development periods the team can deliver functional building blocks periodically to stakeholders who can then provide feedback thus resulting in better project development. The method decreases unexpected delays while guaranteeing that the final system stays focused on user needs. Agile helps teams communicate openly and solve problems quickly through daily meetings, sprint reviews, and retrospectives. These practices ensure that the project stays on track and meets business goals.

Agile development method (Source)

Agile surpasses the conventional Waterfall approaches because it enables continuous modification process-wide without large costs or delays thus serving effectively in flexible development situations. Early detection of functional or technical issues occurs within the multiple development cycles which minimizes project setbacks to produce smoother development processes. Choice fell on Agile due to its efficiency model as well as its emphasis on constant improvement together with a direct focus on delivering valuable results for end-users.

In this project, Agile will guide:

- Frequent user feedback: Ensuring the app meets real-world needs.
- Incremental delivery: Allowing early testing of features like GPS tracking and route management.
- Continuous improvement: Addressing feedback and refining the product throughout its lifecycle.

This methodology creates a dynamic development environment where flexibility and quality and quality are prioritized, leading to a successful project outcome.

The Project development will follow Agile Methodology, focusing on iterative progress and active user feedback.

- Planning: Defines project goals, scope, and timeline.
- Design: Create wireframes and mockups for the app, ensuring alignment with user needs
- Development: Implement features in sprints using react native on mobile for the front end and Node.js and Express.js for the backend.
- Testing: Perform rigorous testing for functionally, usability and compatibility.

3.2 Implementation

3.2.1 Sprint 1

Most of Sprint One was focused on studying the project and outlining how to develop RouteMate. We opened the sprint by focusing on main issues affecting users, for example, they do not know the position or arrival time of their bus. The supervisor and I had frequent meetings to set the target goals for the project and gather valuable details. At this time, the team decided what features the system ought to include, for example, live bus tracking, making bookings and paying online. All the requirements were clearly specified in a document. The app team wrote user stories to show what features users must be able to use in the app such as joining, following buses and booking tickets. It was also split into smaller sections called User Management, Real-Time Transit and Booking. A WBS was used to divide all the project tasks into smaller pieces. It enabled the team to decide on the next steps for the next sprints. The Functional Decomposition Diagram was created to illustrate each step of the system's working process. By the end of Sprint One, everyone on the team understood what the project needed and was set to design its solutions in Sprint Two.

3.2.2 Sprint 2

During Sprint Two, the primary task was to design the user interface and the structure of the system. At first, the team used Figma to design wireframes that helped organize how different parts of the app should be arranged as users navigated from one screen to another. Making the wireframes allowed us to plan the user experience and ensure it remained easy to use. In addition, using Draw.io, I made diagrams such as use case, sequence and entity relationship diagrams (ERD). They gave the team an idea of how the units in the system would interact. All user, booking and bus data were secured in this sprint by designing the database with PostgreSQL. Reviewing each design with the supervisor allowed the team to improve it based on their comments. Because of the sprint, the team could settle on the system architecture that would lead development for the rest of the sprints. Because the planning was so clear in this Sprint, writing code in Sprint Three was not difficult.

3.2.3 Sprint 3

During Sprint Three, the main features were developed. For this stage, they created the User Management features which allow users to register, log in and convert between hands of passenger or driver. The app was built using React Native, so it can run on both Android and iOS. On the server side, work was done using Node.js and Express.js to manage login, user

data and messages sent by the server. All our data is stored within the PostgreSQL database. Validating logins and option to reset the password are included in the system. The introduction of “Driver Mode” allows users to upload their papers and decide on their car. The code was managed and its changes tracked using GitHub. The goal for this sprint was to allow users to join, log in and manage their account safely. At the close of this sprint, the main structure of the app was created and put through testing, so new features could be developed in the following sprint.

3.2.4 Sprint 4

In Sprint Four, time was spent on creating features for real-time tracking and for booking. They used Socket.io to allow drivers to send their locations and let users view them live on a map. With Leaflet and OpenStreetMap, the map displayed to users the location of the current bus and how long until arrival. Due to this feature, passengers could now arrange their trips much simpler. A Booking Management System was created as well so people could book seats, select their buses and set their travel time. People using the app were allowed to cancel their trips at least 20 minutes prior to departure. Administrators could watch all reservations and drivers could look at the list of passengers. Users were notified by the app after making a new reservation. This sprint was essential because the basic tools users needed every day were added. We also validated that the features functioned well while processing real-time data.

3.2.5 Sprint 5

During Sprint Five, the team finished developing the remaining features and focused on analyzing the system. The major job involved linking Khalti and eSewa for digital transactions. As a result, tickets could be bought safely by Khalti. It managed the safety of transactions and stated the status of each payment for every user. Additionally, passengers can now rate their trip using the Rating and Review System. Reviews could be analyzed by admins to improve the company’s service. As soon as the features were complete, the team started checking the APIs by running tests with services like Postman. A test case was designed for each of the main actions like login, tracking tickets and payment. The app was updated after users provided their opinions. Every planned feature was running well by the time I finished my work. Users could now use RouteMate to track their trip, arrange the booking and pay with ease.

3.3 Work Breakdown Structure

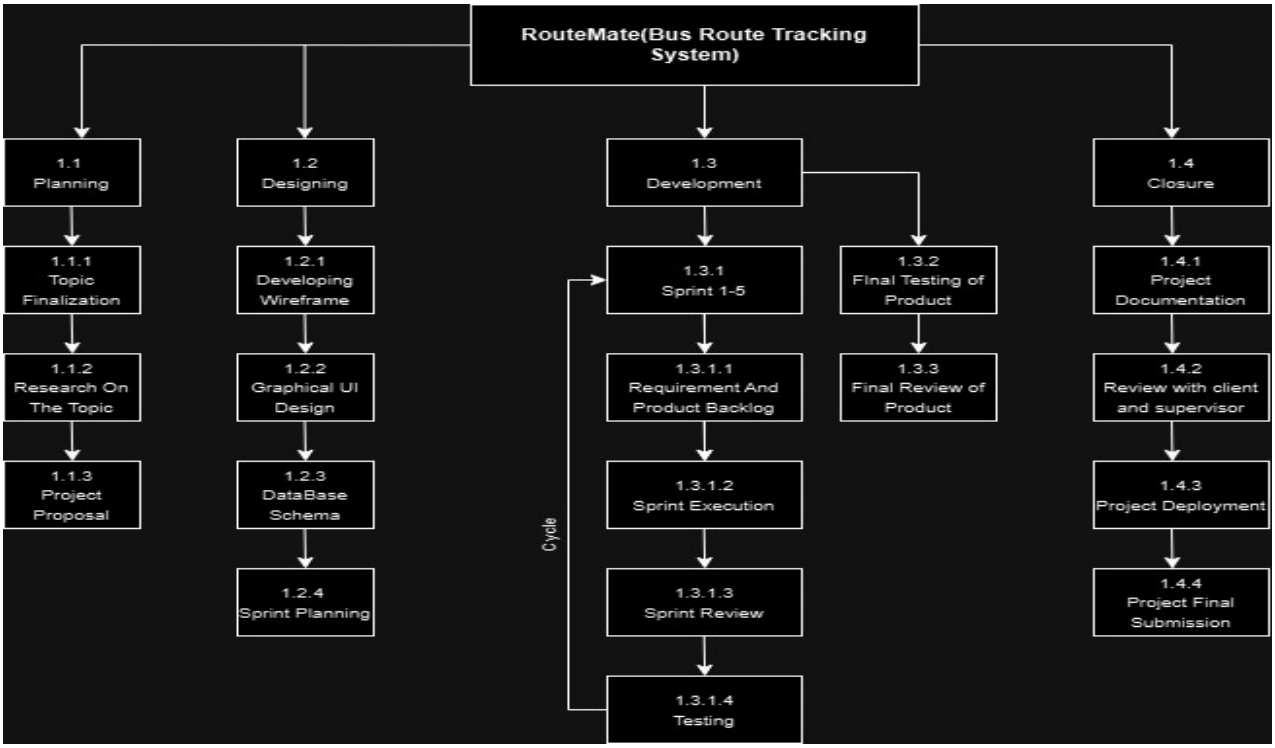


Figure 7: Work Breakdown Structure

4. Tools and Technology

4.1 Figma

I plan to design wireframes in Figma, which will serve as the blueprint for the UI of the Bus Route Tracking System (RouteMate). This Step is crucial for aligning the design with user needs and system requirements.

4.2 Draw.io

It was used to make detailed diagrams to understand, manage the projects and also to make use case diagrams, sequence diagrams, activity diagrams and class diagrams needed for developing artefacts.

4.3 Front End

- React Native: For developing the cross-platform mobile application.
- UI kits and Styles: For developing better User Interface

4.4 Backend Development

- Node.js: For building the server-side application
- Express.js: To handle routing and API development.
- Socket.io: For enabling scalable and efficient real-time updates.

4.5 Maps and API

Using Leaflet with OpenStreetMap and its API includes geolocation features, the option to customize the map and APIs are used to fetch the bus stops and for routing.

4.6 Khalti Integration

Online transactions are safe, quick and convenient, due to the use of Khalti Payment API its focus on encrypted payments and available options, the payment process on this gateway becomes more reliable and straightforward.

4.7 Database Management

- PostgreSQL: For database management, user authentication, and cloud storage.

4.8 Version Control

- Git: GitHub for code repository and collaboration.

4.9 Testing

- Postman: For API testing and ensuring proper backend functionality.

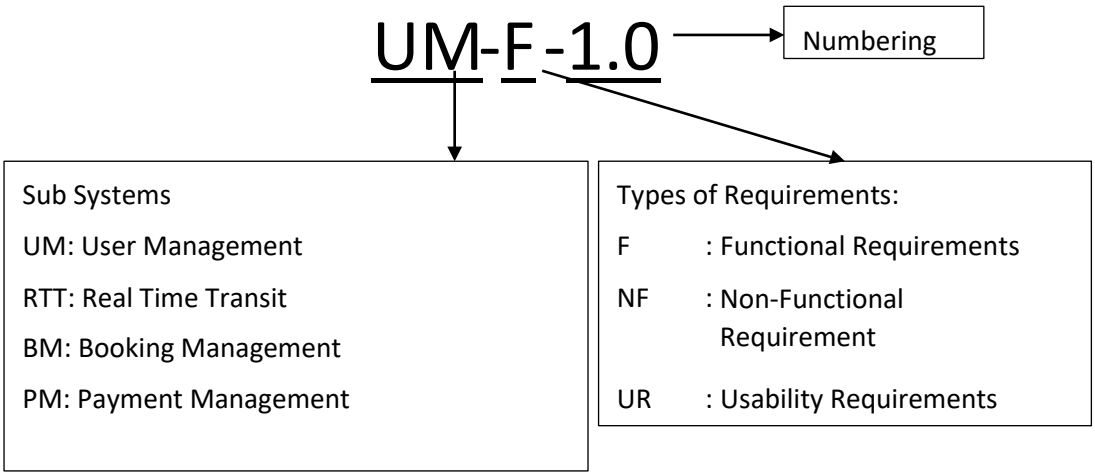
4.10 Deployment

- Google Play Store for publishing the Android application in near future.

5. Artefact design

5.1 Software Requirements Specification

5.1.1 Legend



5.1.2 User Management

Requirement Code	Requirement Description	Use Case
UM-F-1.0	The platform must enable new users to create an account using a distinctive username, a secure password, and a legitimate email address.	Register User
UM-NF-1.1	The signup password should contain a minimum of 8 characters and feature a combination of alphabets, digits, and special characters.	
UM-UR-1.2	Users should provide the option to show or hide their password	
UM-UR-1.3	User should have the ability to reset their password if they forget it.	
UM-F-2.0	After successfully registering, The user is navigated to the login page.	
UM-UR-2.2	The user should be notified of a successful registration and redirected to a new page.	
UM-F-3	User can to switch to Driver mode using the role switch Button.	
UM-F-3.1	User can navigated to document verification page when the button is clicked.	
UM-F-3.2	Users are allowed to choose the vehicle option according to users preference	

5.1.2.1 Use case Diagram of User Management

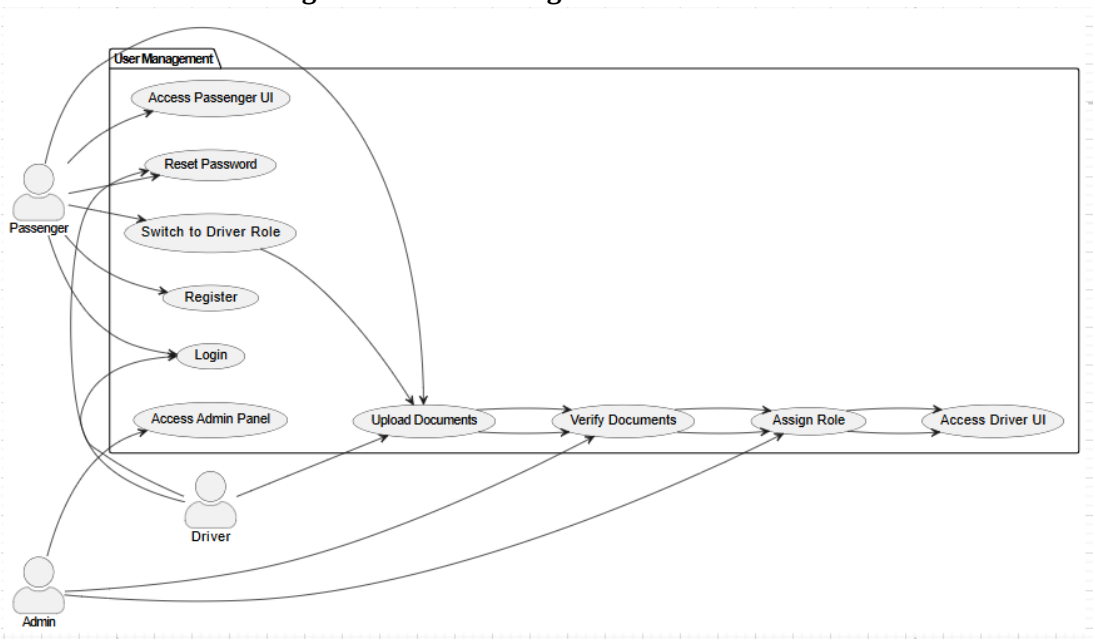


Figure 8: Use case diagram for User management

5.1.2.2 Sequence Diagram of User Register

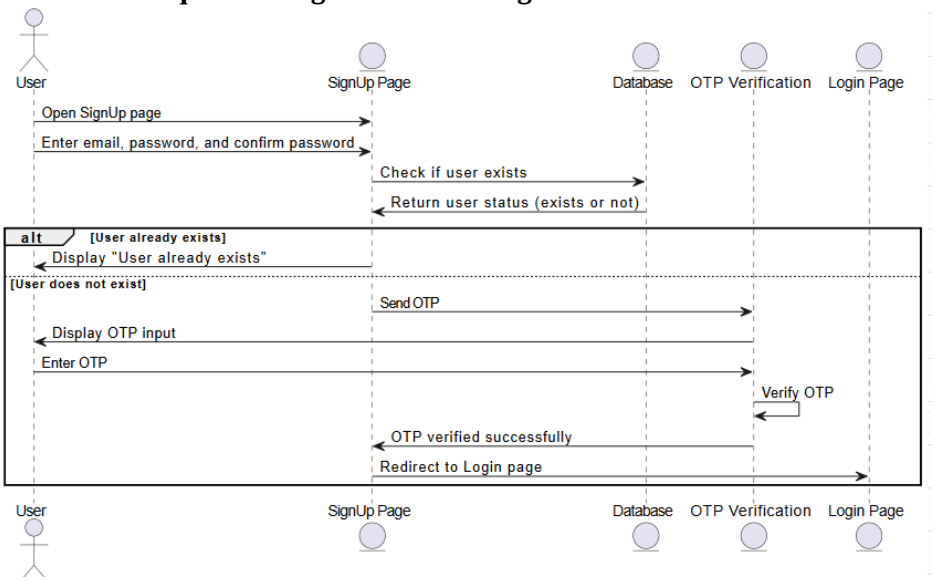


Figure 9: Sequence Diagram for User Register

5.1.2.3 Sequence Diagram of User Login

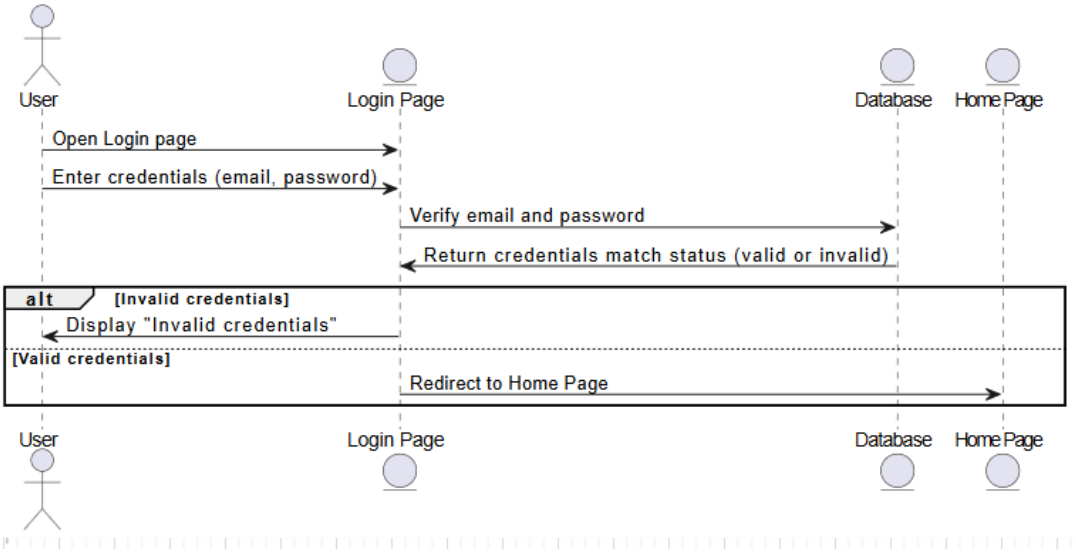


Figure 10: Sequence Diagram for User login

5.1.3 Real-time Transit System

Requirement code	Requirement Description	Use case
RT-TS-1.0	The system allows both admin and users to see the bus location	
RT-TS-1.1	The Driver should go online in order to show the bus location using real time data	
RT-TS-1.2	The system provides real-time updates of bus locations, arrivals, and departures, allowing users to track buses along their routes.	Track Bus in Real-Time
RT-TS-1.3	The system should show live bus locations on a map in real-time, allowing users to see the current position of buses on their selected routes.	
RT-TS-1.4	Users can view the estimated time of arrival (ETA) for buses and distance for preferred location.	

5.1.3.1 Use case Diagram of Real time Transit

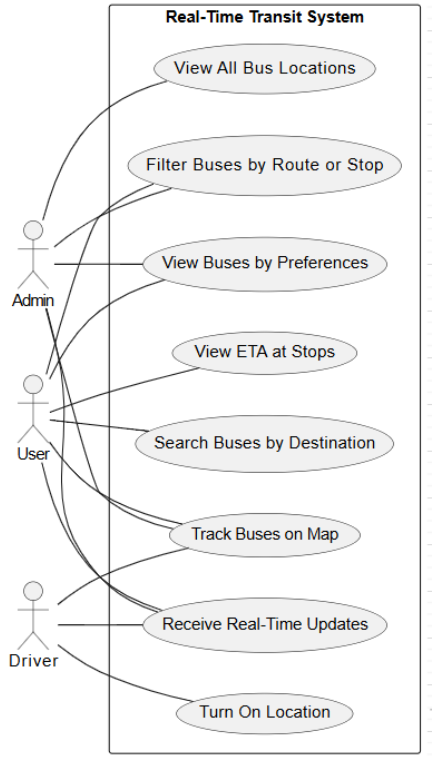


Figure 11: Use case Diagram of Real time transit

5.1.3.2 Use case Diagram of Real time Transit

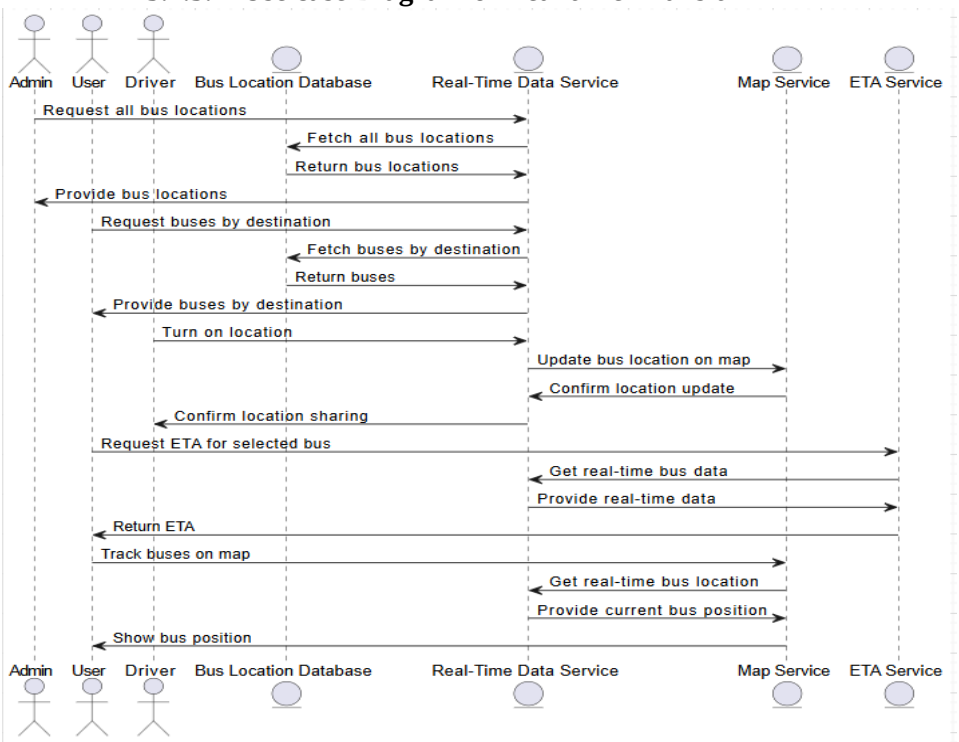


Figure 12: Real time transit sequence diagram

5.1.4 Booking Management System

Requirement code	Requirement Description	Use case
BM-F-1.0	Users can book tickets by selecting a location, bus, seat, number of tickets, and exact date/time.	Book Ticket
BM-NF-1.1	Error messages are shown if required fields are missing.	
BM-UR-1.2	After booking, users receive a confirmation message with ticket details (bus number, tickets, date and time).	
BM-F-2.0	Booking history can be viewed by users, including current, cancelled, and previous bookings.	
BM-F-3.0	Booking can be cancel up to 20 minutes before bus departure. No refund for cancellations made within 20 minutes.	
BM-NF-3.1	The system notifies users upon booking cancellation.	
BM-UR-3.2	Users are informed that refunds may take a few days.	
BM-F-4.0	Admins can view all bookings, including user details, bus number, tickets, date, and time.	Admin Viewing All Bookings
BM-F-5.0	Drivers can view passenger details for their bus, including booking information.	

5.1.4.1 Use case diagram for Booking Management

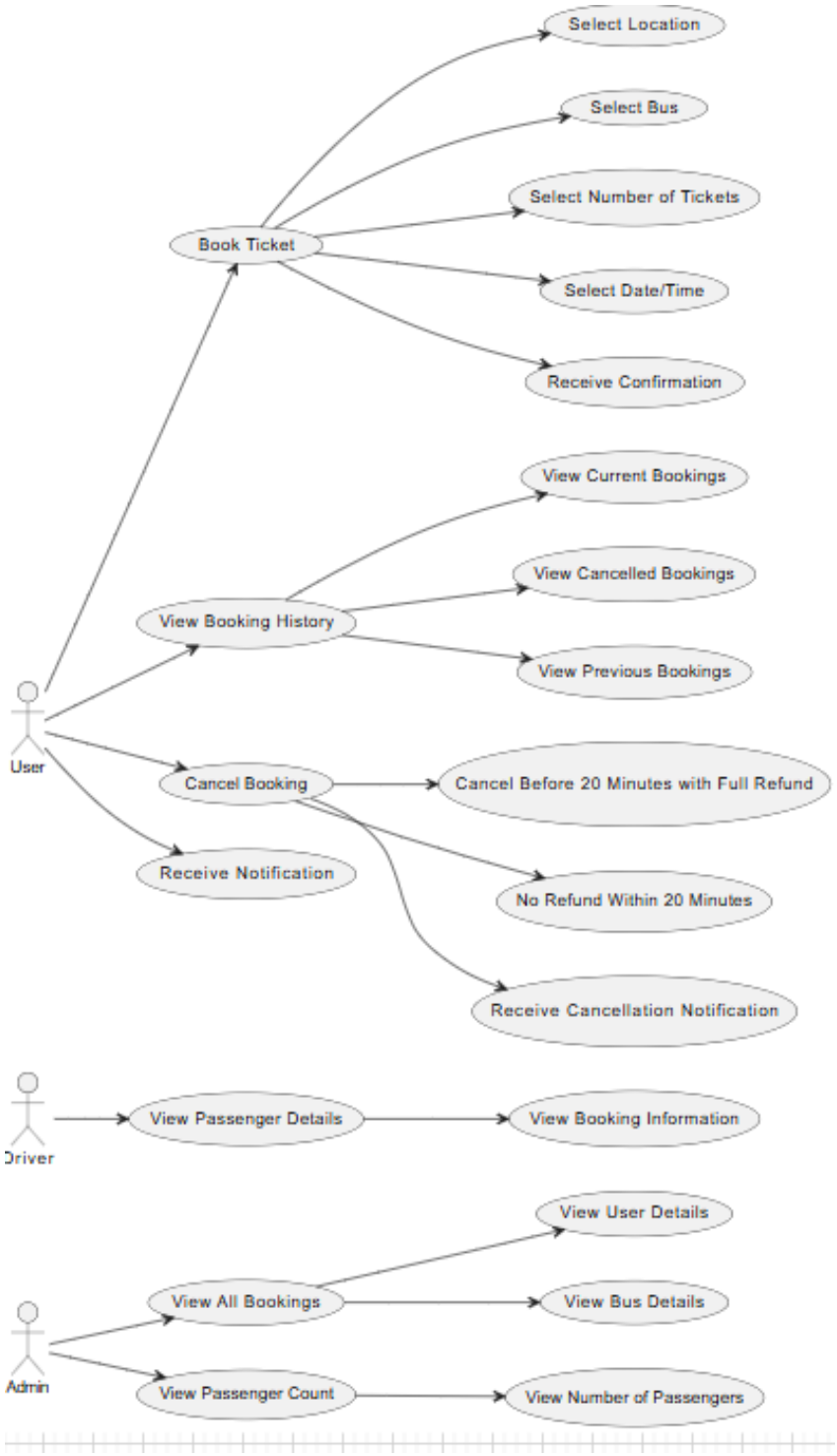


Figure 13: Use case for booking management

5.1.4.2 Sequence Diagram for Booking Management System

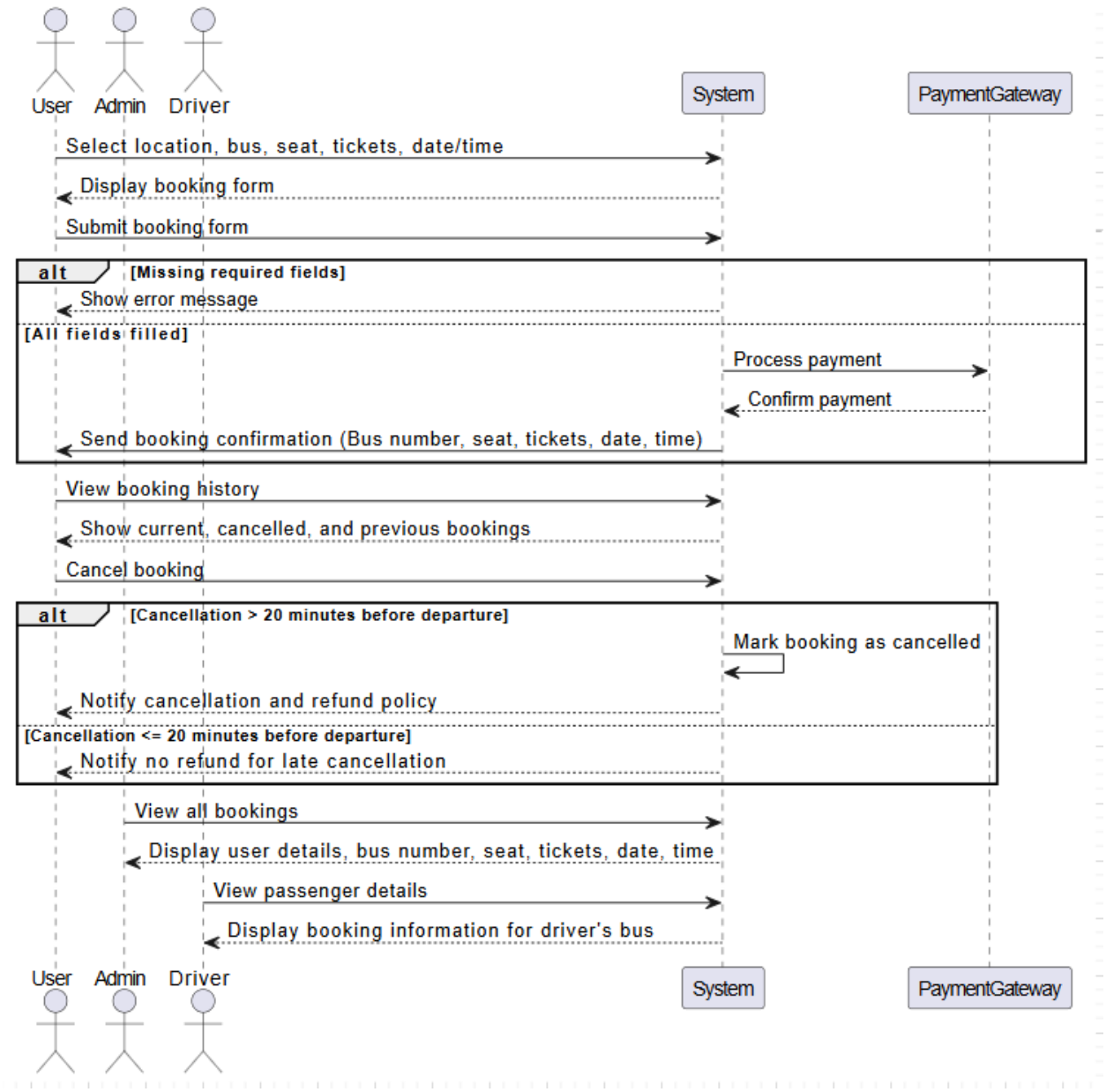


Figure 14: Sequence Diagram for Booking

5.1.5 Payment Management System

Requirement Code	Requirement Description	Use Case
PM-F-1.0	The system offers various payments integration, including esewa or Khalti.	Payment
PM-F-1.1	The payment system supports contactless methods.	
PM-F-1.2	The payment process are carried out securely through an encrypted gateway.	
PM-F-1.2	Users are notified regarding the outcome of the payment, whether it is successful or unsuccessful.	
PM-F-2.0	Both Admin and users should have access to payment history.	
PM-F-2.1	Users can check the payment status of all their booked tickets.	

5.1.5.1 Use case diagram for Payment Management system

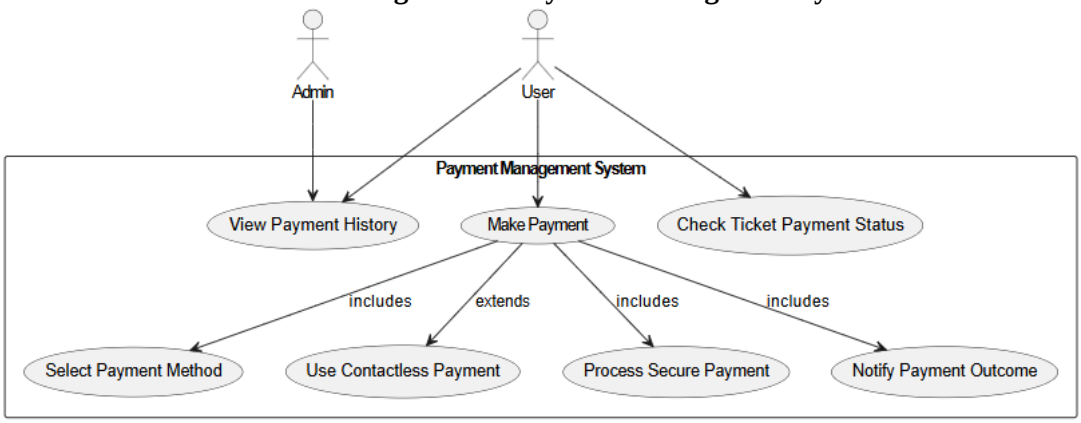


Figure 15: Use case diagram for Payment Management system

5.1.5.2 Sequence Diagram for Payment Management system

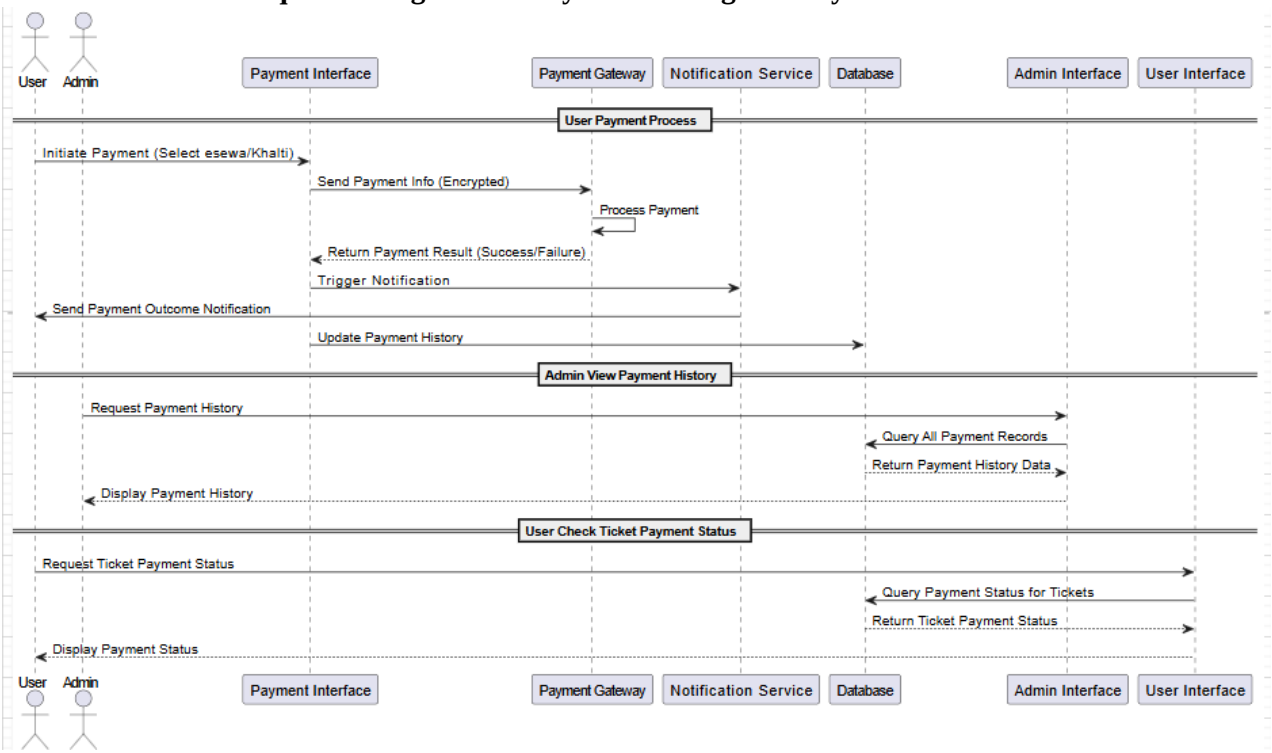


Figure 16: Sequence Diagram for Payment Management system

5.1.6 Rating and Review System

Requirement Code	Requirement Description	Use Case
RR-001	User can submit a rating and text review	
RR-002	Displays driver rating in while selecting the bus in passenger panel	
RR-003	Users can edit or delete their own reviews	
RR-004	Admin can view all the reviews and ratings	

5.1.6.1 Use case diagram for Rating and review system

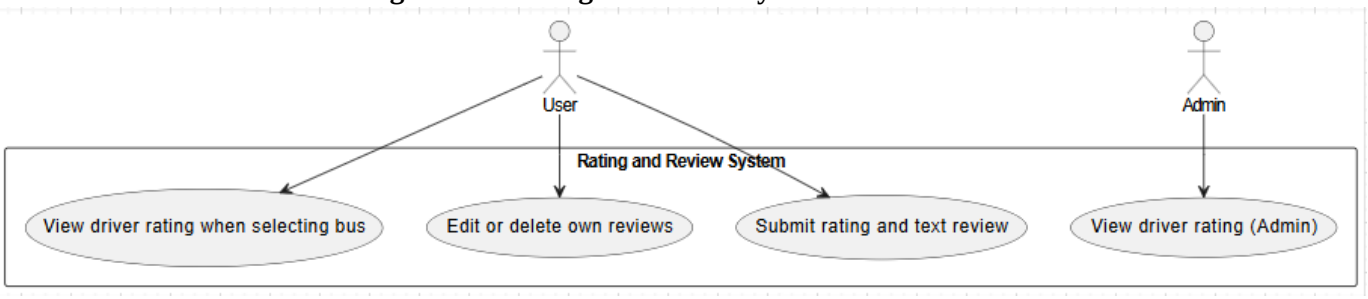


Figure 17: Use case diagram for rating and review system:

5.1.6.2 Sequence diagram for Rating and review system

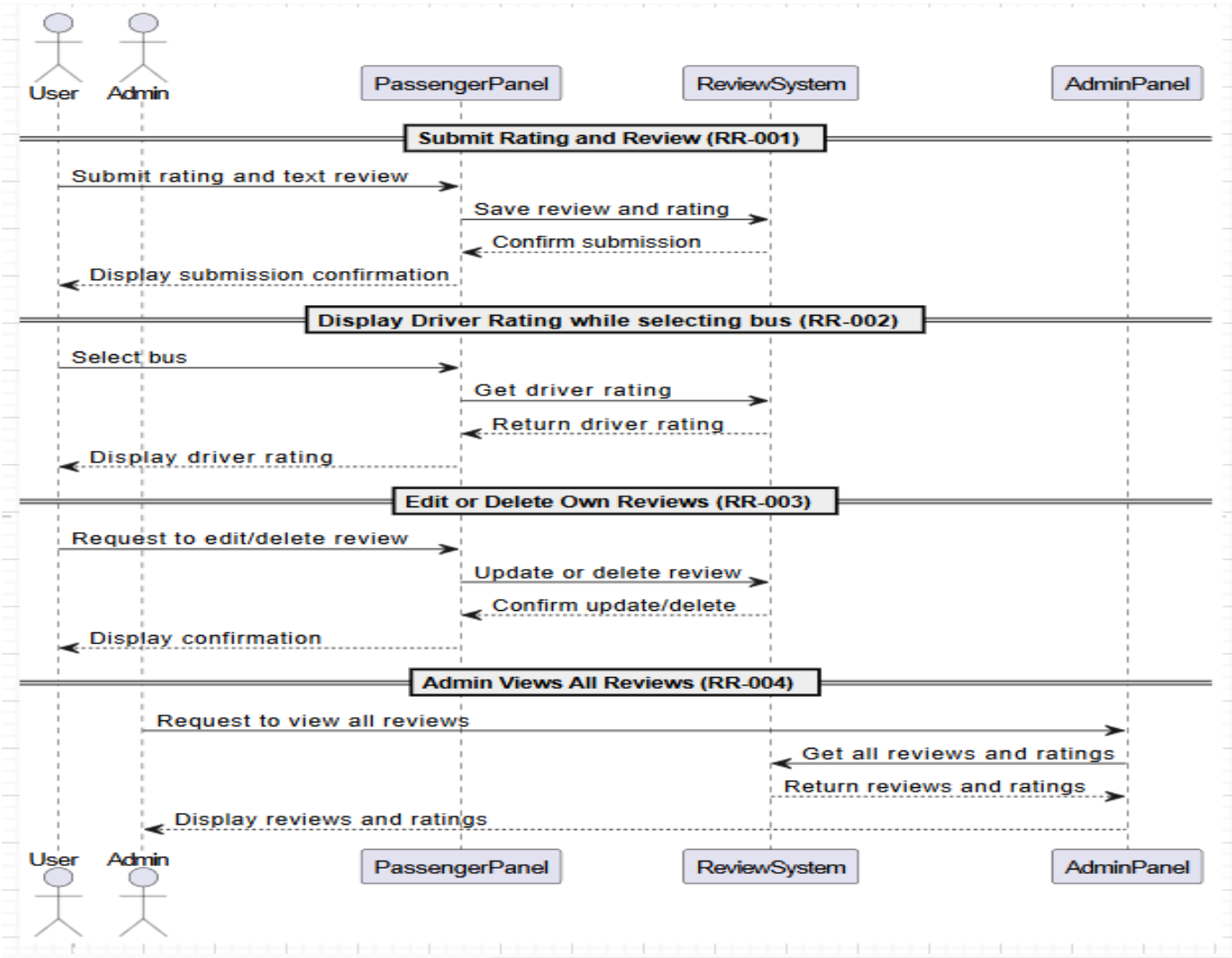


Figure 18: Sequence diagram for rating and review system

5.1.7 Entity Relationship Diagram

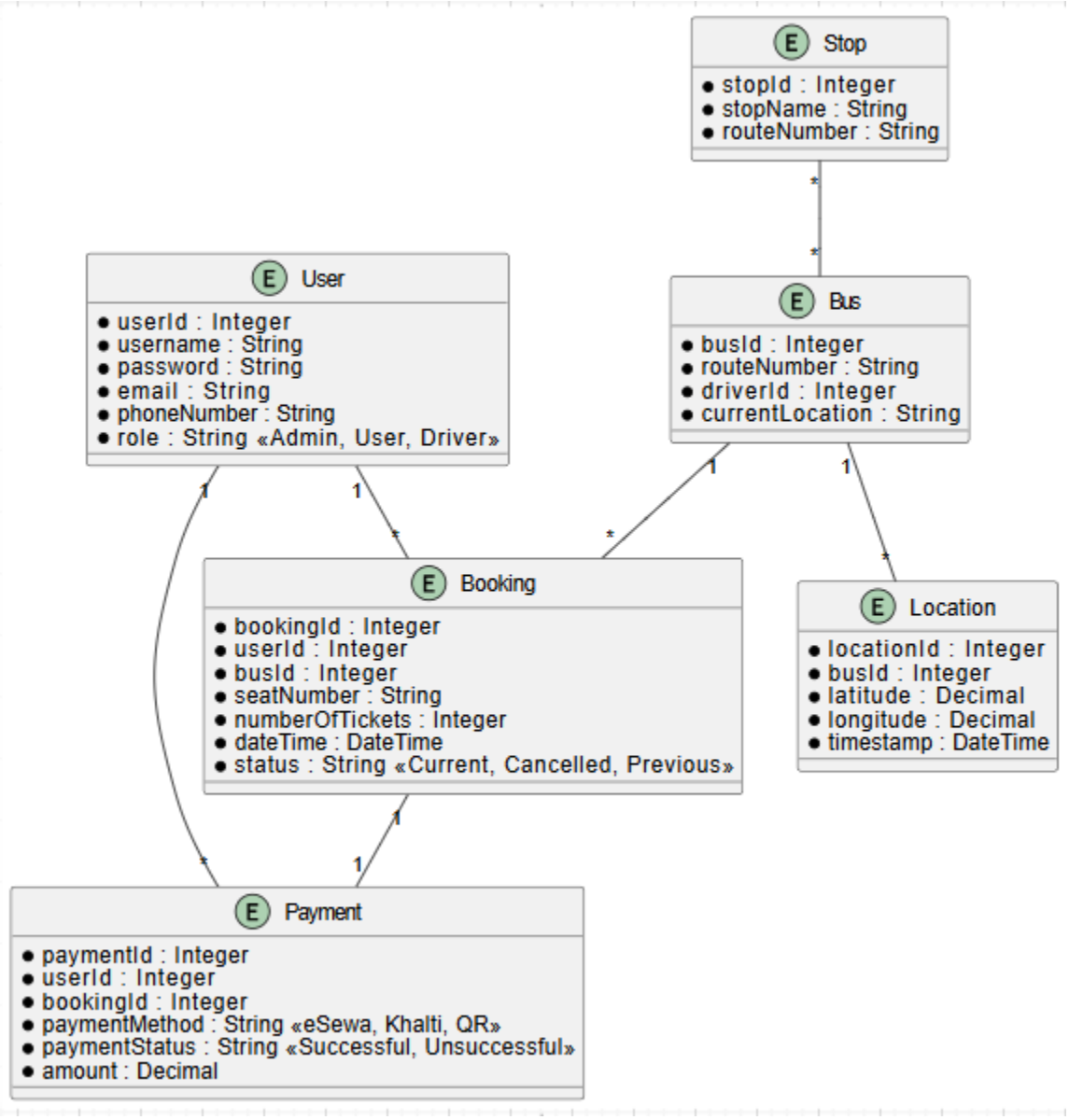


Figure 19: Entity Relationship diagram

5.2 Test Case

5.2.1 User Register

Test Case	TC_01
Objective	To register a new user with unique email during registration.
Action	Register new user
Expected Results	The new user has completed the registration successfully.
Actual Results	Registration completed successfully.
Conclusion	Success

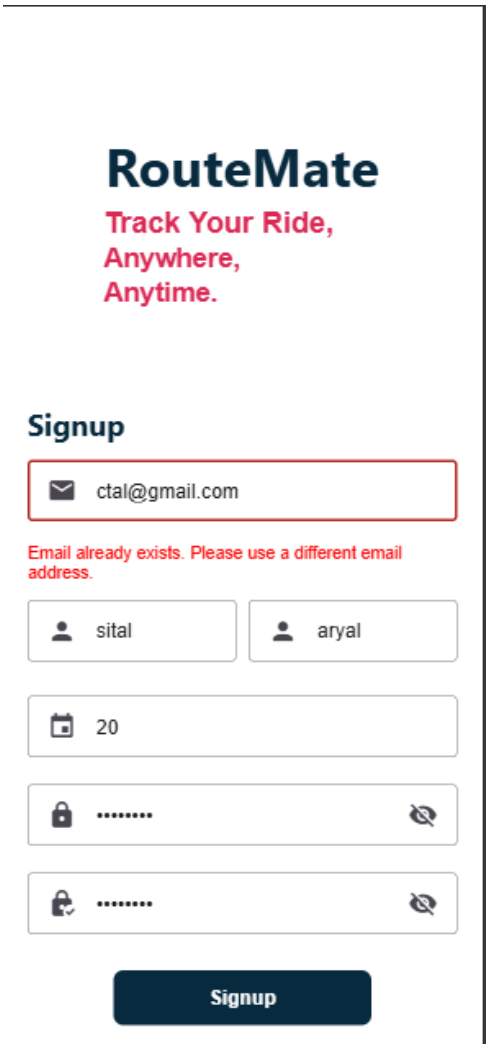


Figure 20: Email already exist message

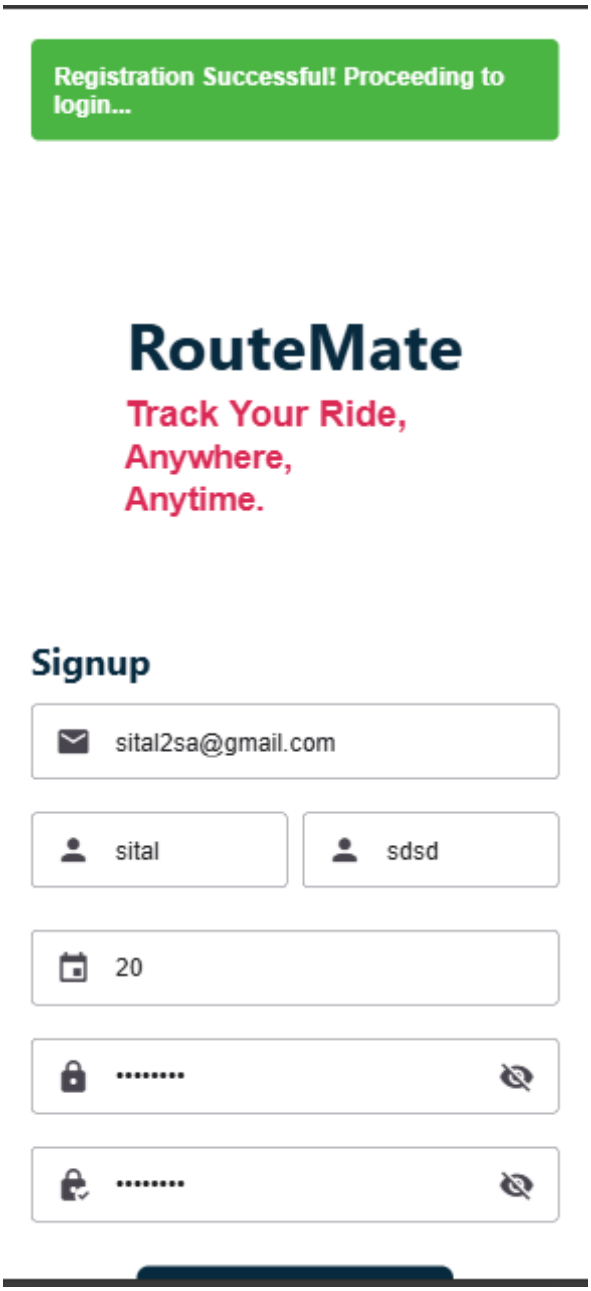


Figure 21: Registration successful message

5.2.2 User Login

Test Case	TC_02
Objective	To test if existing user can login on the app.
Action	Login existing users
Expected Results	To check if the user is already register and procced to login
Actual Results	Login successfully and redirects to home page
Conclusion	Success

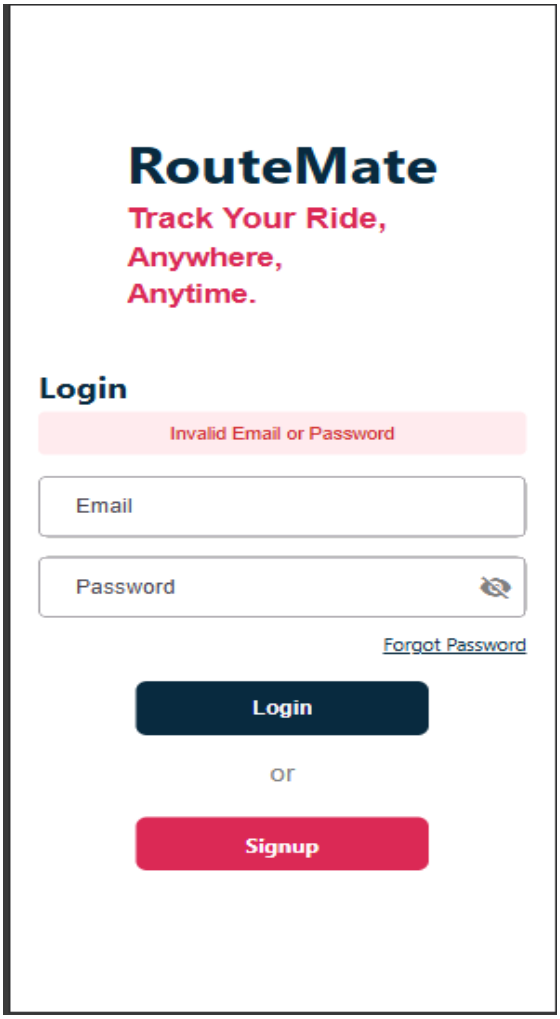


Figure 22: Invalid Email or password image

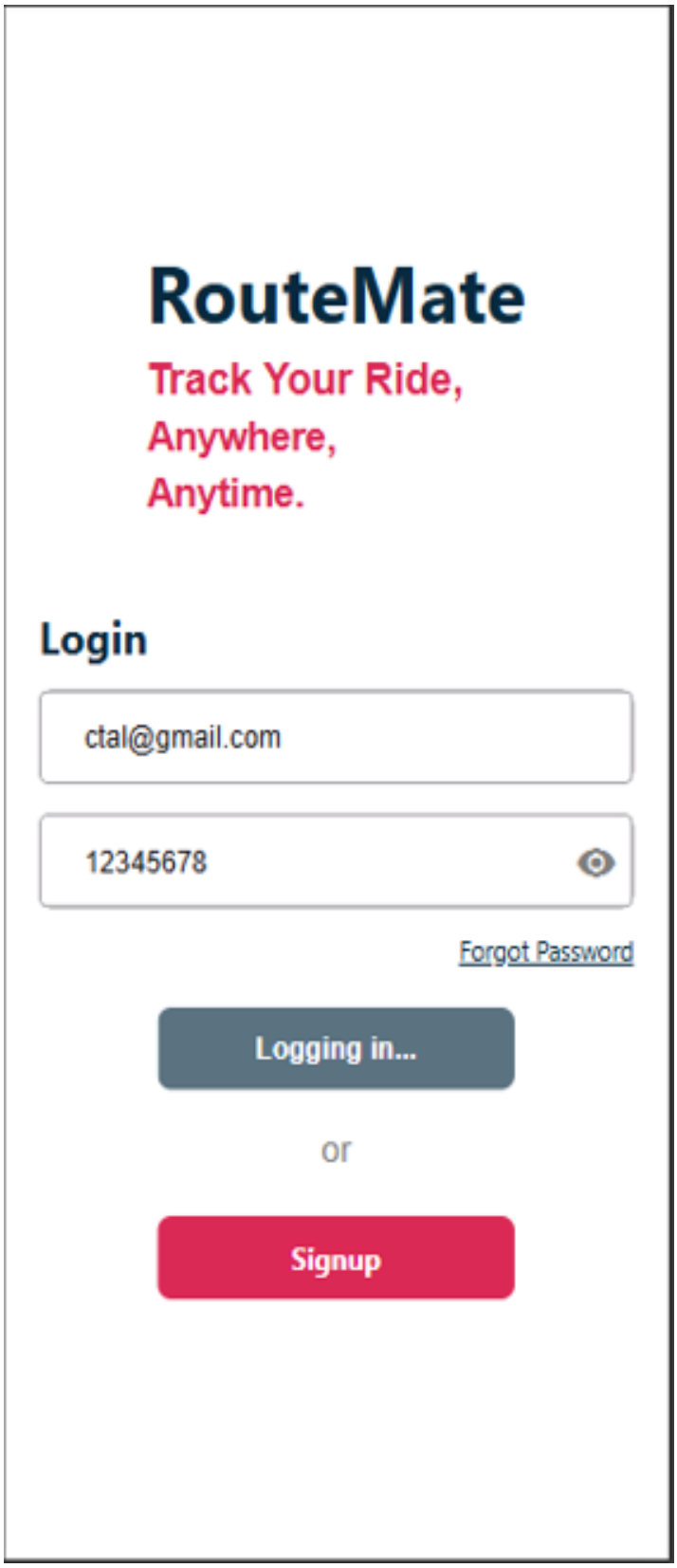


Figure 23: Successful login image

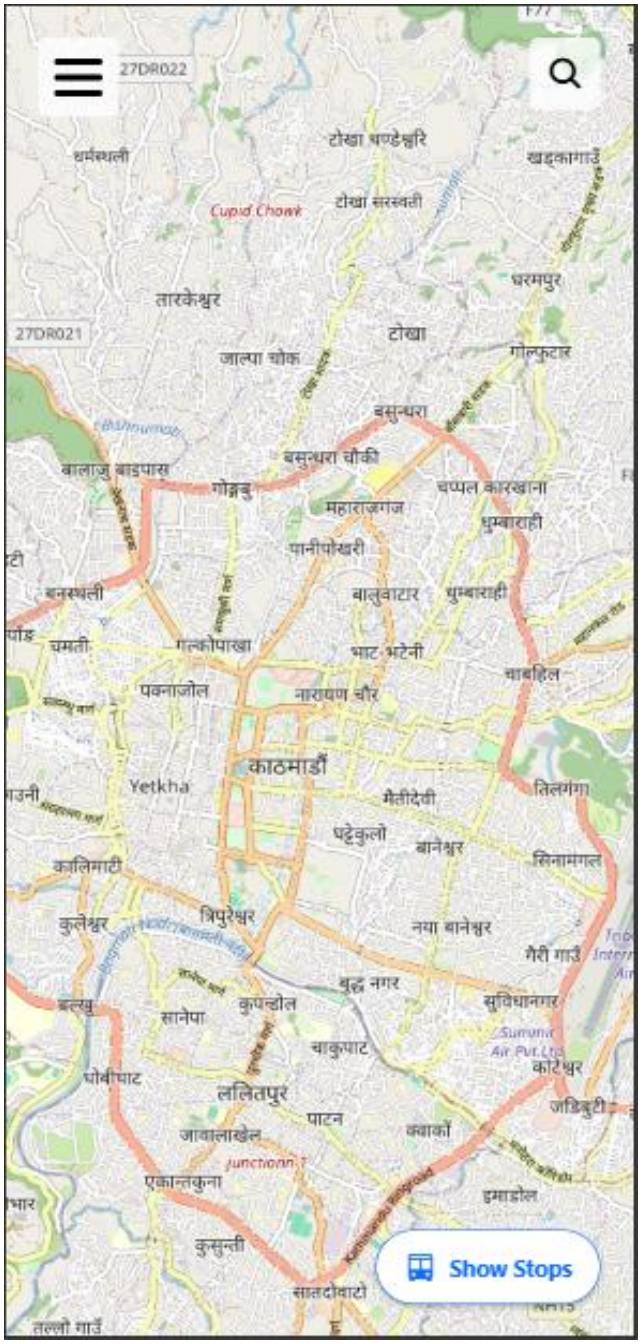


Figure 24:Redirects to home page after login

5.2.3 Forget Password

Test Case	TC_03
Objective	To check if the user can change to new password if they forget it
Action	Set new password
Expected Results	To check if the user can set new password after OTP verification
Actual Results	New password set
Conclusion	Success

5.2.4 Logout

Test Case	TC_04
Objective	To check log-out functionality.
Action	Click the logout button available on the hamburger menu.
Expected Results	The App should log out the user and navigates back to the login.
Actual Results	Log out successfully and redirects to Login page
Conclusion	Success

5.2.5 Map integration

Test Case	TC_07
Objective	To see if the mark is integrated on the home page
Action	Map integrated
Expected Results	Map of the Kathmandu should be shown
Actual Results	Map shown
Conclusion	Success

Figure 25: Map integration using open street map

5.2.6 Role switch button

Test Case	TC_06
Objective	To check the role switch functionality.
Action	Click the role switch button available on the hamburger menu.
Expected Results	The button should take use to document upload page or if already verified the takes user to driver page.
Actual Results	Takes user to document upload page for document Verification or if already have a driver role user goes to drive panel
Conclusion	Success

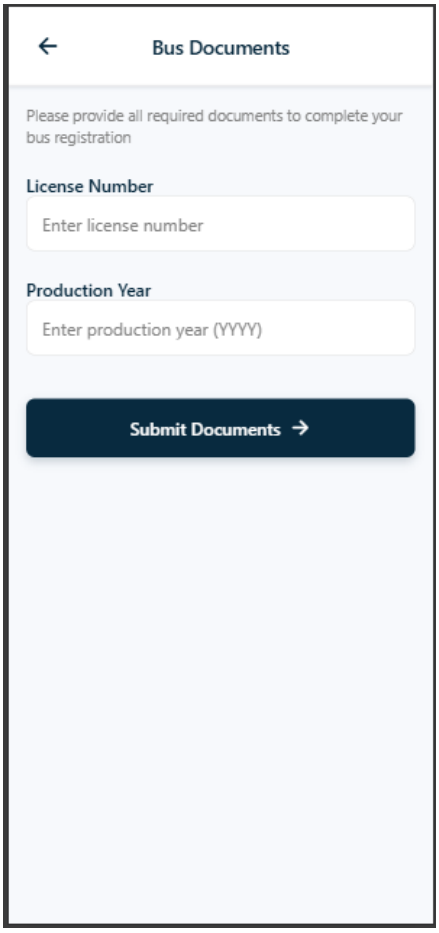


Figure 26: Document Upload page for driver verification

Figure 27: Driver panel after verification

5.2.7 Displays bus live location

Test Case	TC_07
Objective	To see the bus live location by user and admin
Action	Real time location of active buses
Expected Results	Bus should be shown on the map in admin and user panel
Actual Results	Location shown
Conclusion	Success

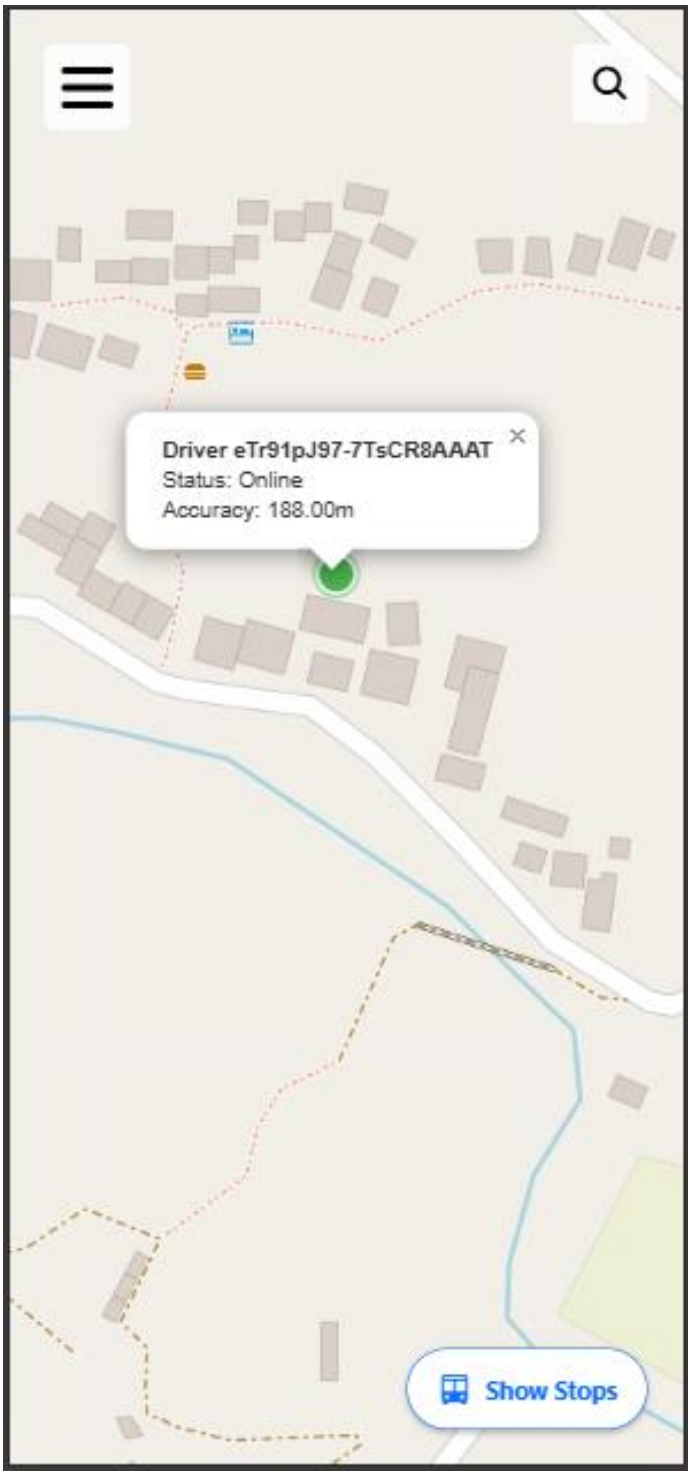


Figure 28: Real time location after driver

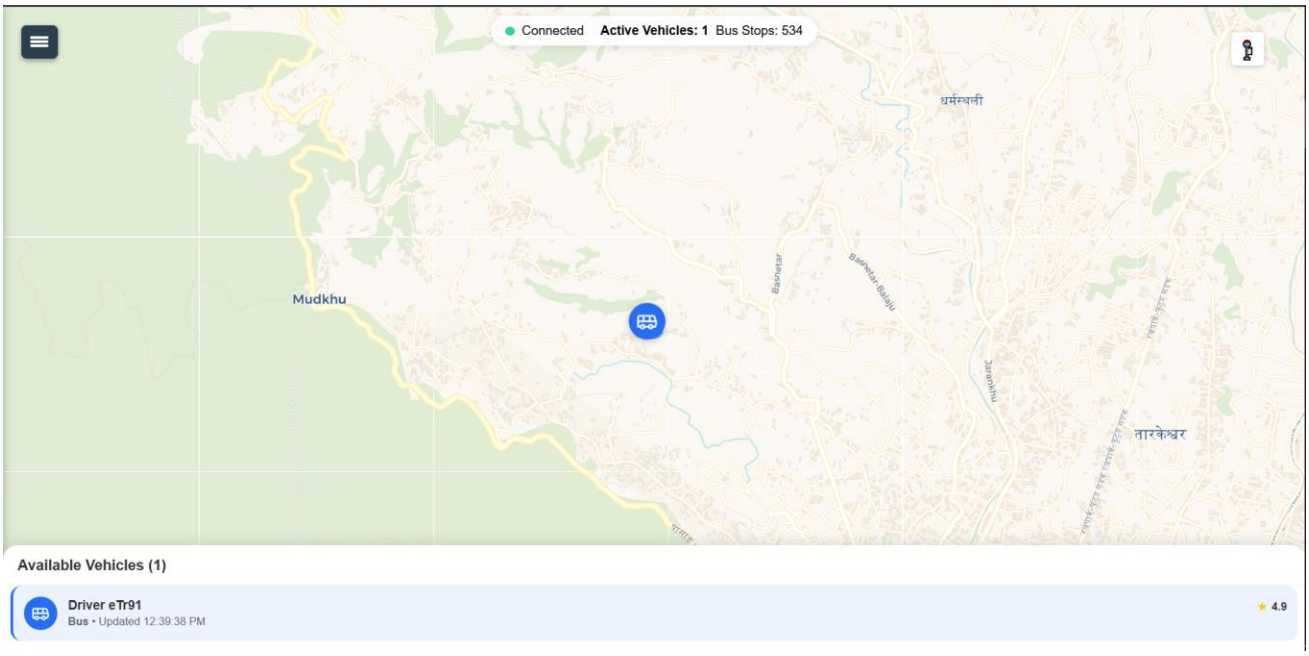


Figure 29: Driver location displayed in Admin panel

5.2.8 Bus Stops

Test Case	TC_05
Objective	To check whether the bus stops are displayed or not on button click
Action	Bus stops are shown and displayed in user panel
Expected Results	The App should show the all the located bus stops on map
Actual Results	Bus stop shown
Conclusion	Success

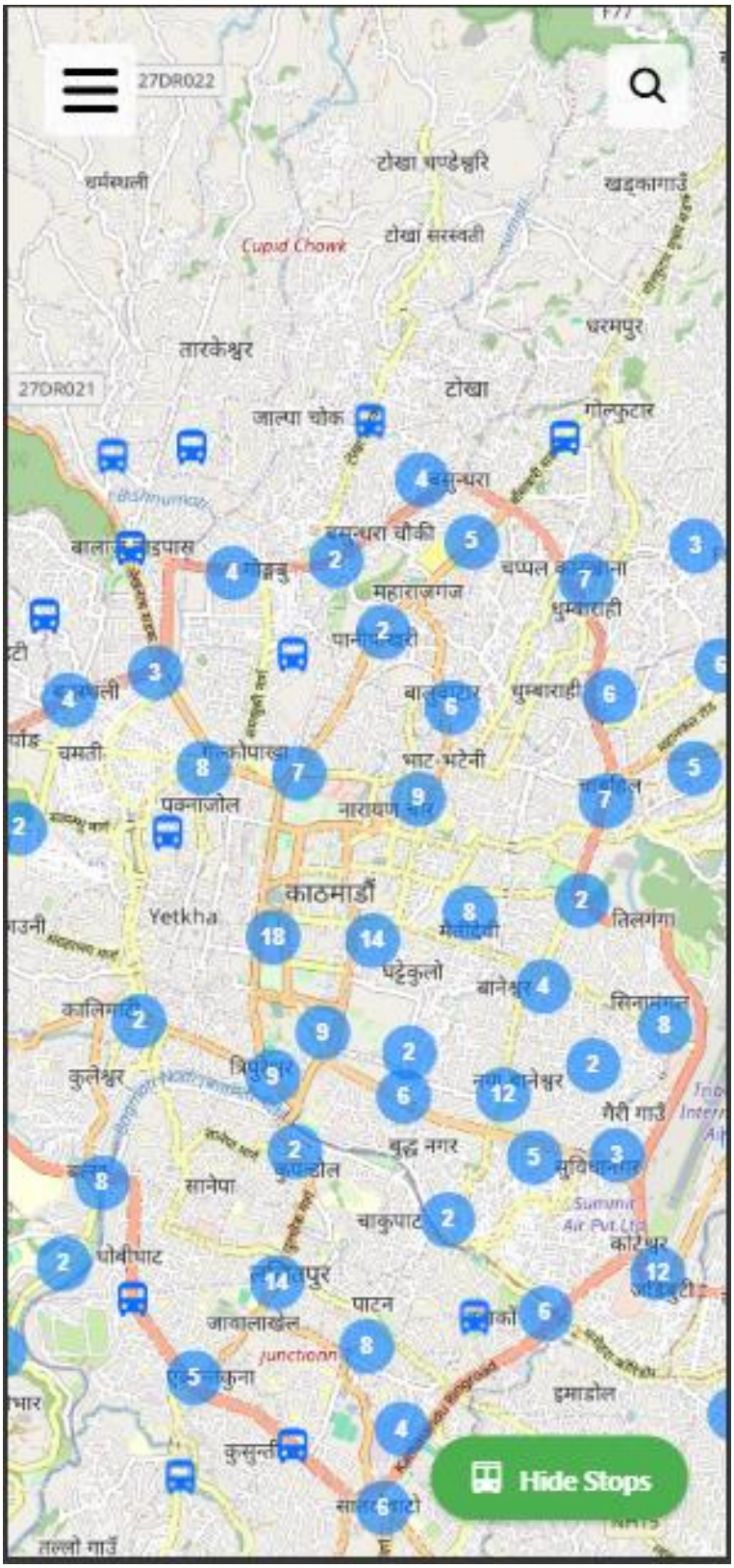


Figure 30: Bust stop shown on show stops button click

5.2.9 Driver Location

Test Case	TC_06
Objective	To check whether if the driver live location displayed or not while online
Action	Location is displayed in the map
Expected Results	The App should show the all the bus location on online mode
Actual Results	Location enabled and displayed
Conclusion	Success

Figure 31: Location displayed of driver on online mode

5.2.10 Route Selection

Test Case	TC_07
Objective	To check if the route is displayed or not from point A to point B
Action	Display route according to user preference
Expected Results	The App should show the route of selected location
Actual Results	Route shown on the map
Conclusion	Success

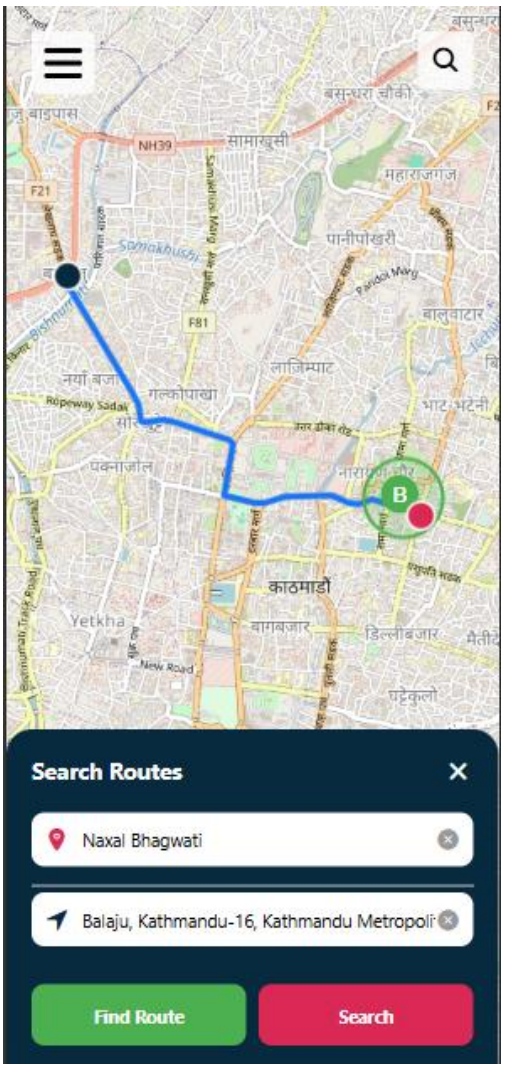


Figure 32: Route displayed on user preference

5.2.11 Estimate Time Arrivals (ETAs)

Test Case	TC_08
Objective	To show estimate arrival time and duration with the help routing.
Action	Display ETAs for preferred location
Expected Results	The App should show expected time for selected location
Actual Results	Estimated time shown
Conclusion	Success

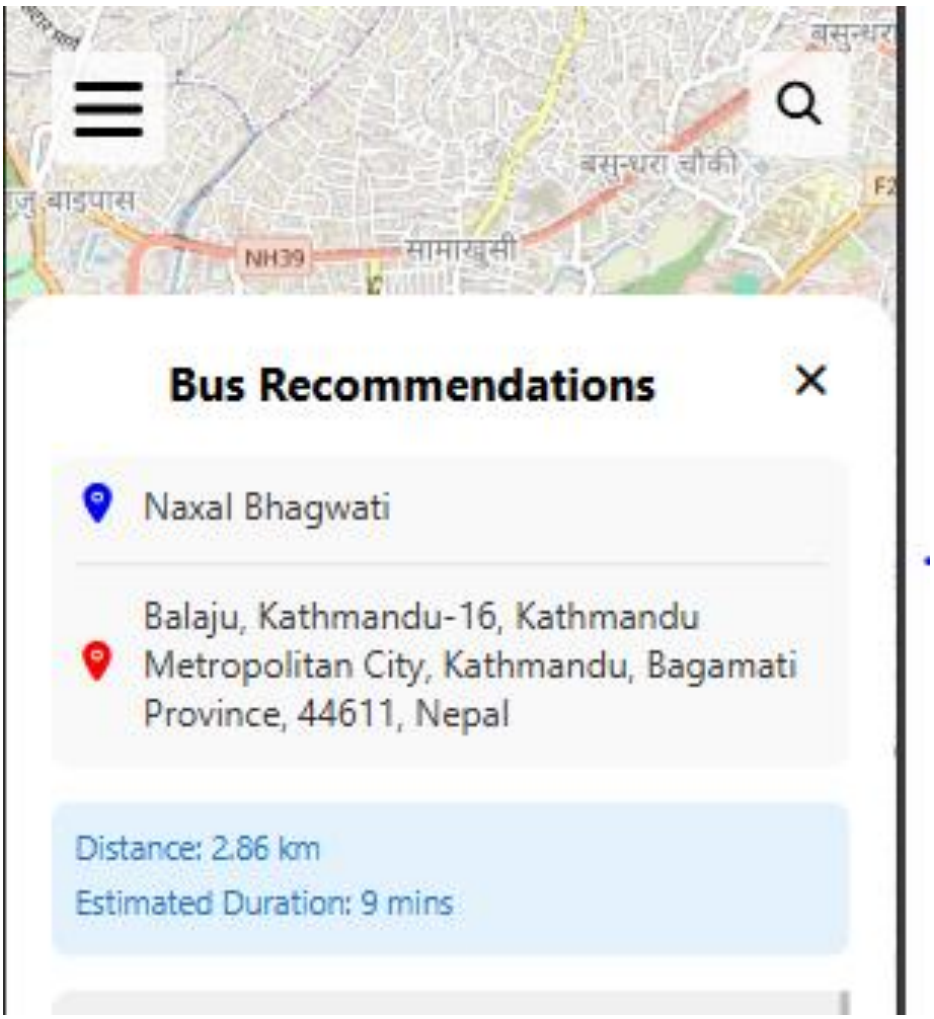


Figure 33: Displayed Estimate time and distance

5.2.12 Ticket Booking

Test Case	TC_08
Objective	To book tickets with user preference.
Action	To successfully book bus ticket.
Expected Results	Ticket booking and after booking it's displayed in booking section.
Actual Results	Successfully book a ticket
Conclusion	Success

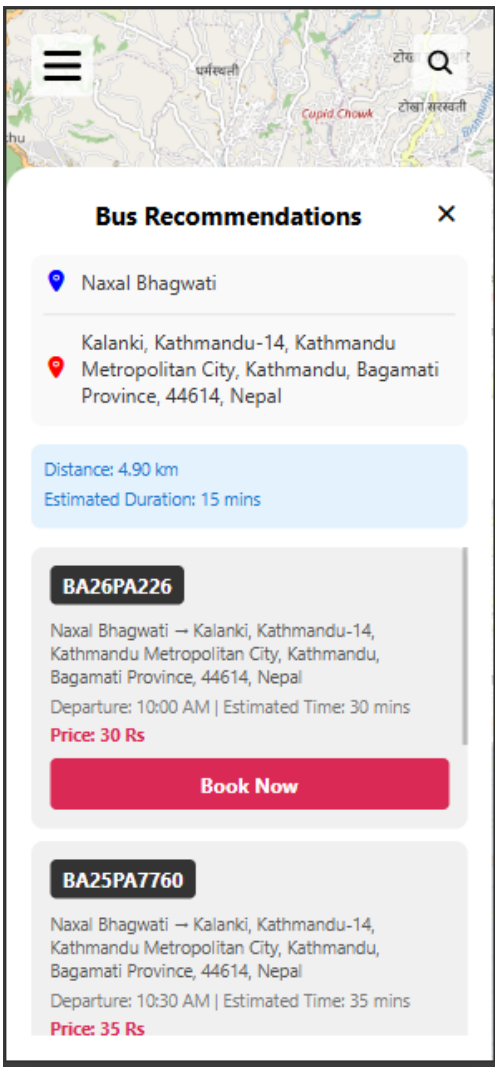


Figure 34: Ticket Booking Details page

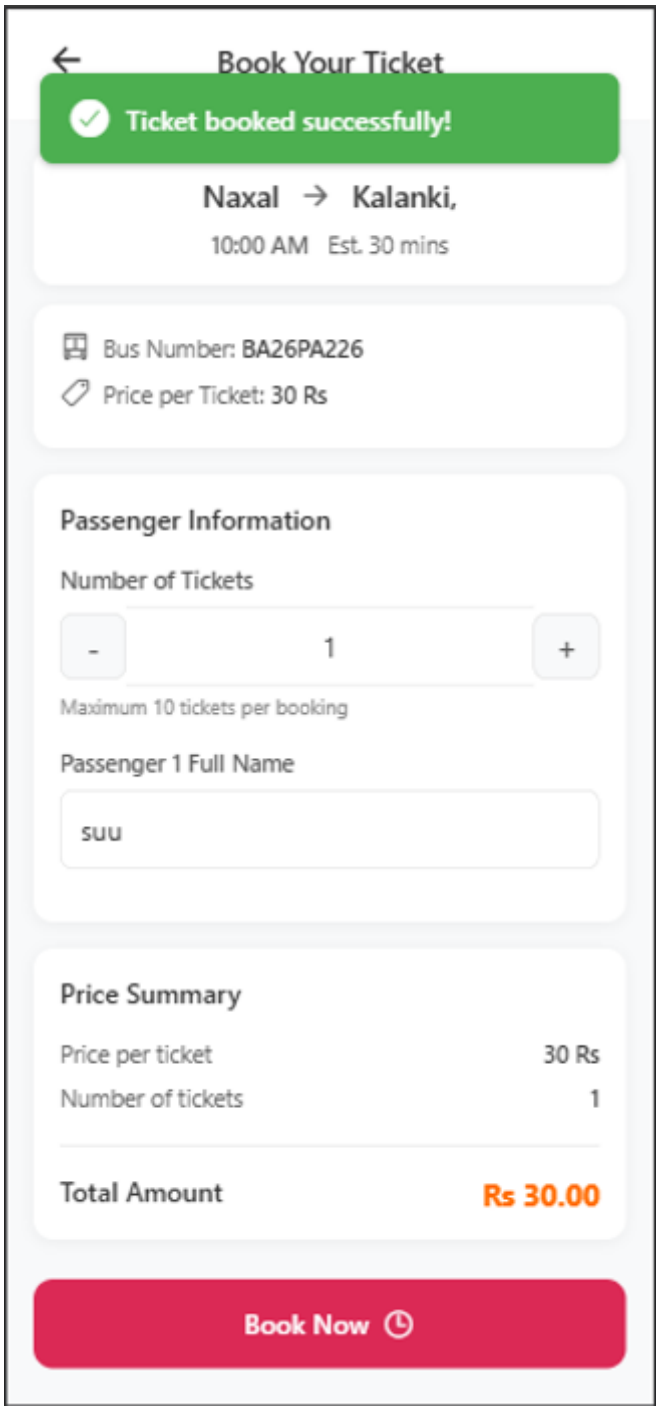


Figure 35: Ticket booked successfully

5.2.13 Cancel Booking

Test Case	TC_08
Objective	To cancel booking of a user within certain timeframe
Action	To successfully cancel bus ticket.
Expected Results	Cancel booking and after cancelled it's displayed in cancelled booking section.
Actual Results	Successfully cancel a ticket
Conclusion	Success

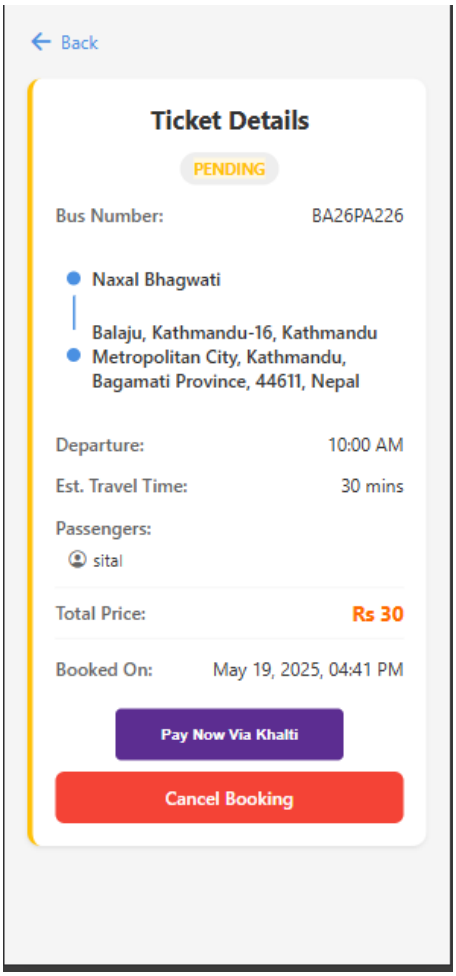


Figure 36: Cancel Booking image

5.2.14 Manage Booking by Admin

Test Case	TC_08
Objective	Admin can view all the booking details and manage it.
Action	To successfully view booking details of all the users
Expected Results	Ticket management and with better UI for admin
Actual Results	Successfully displayed all the booked tickets
Conclusion	Success

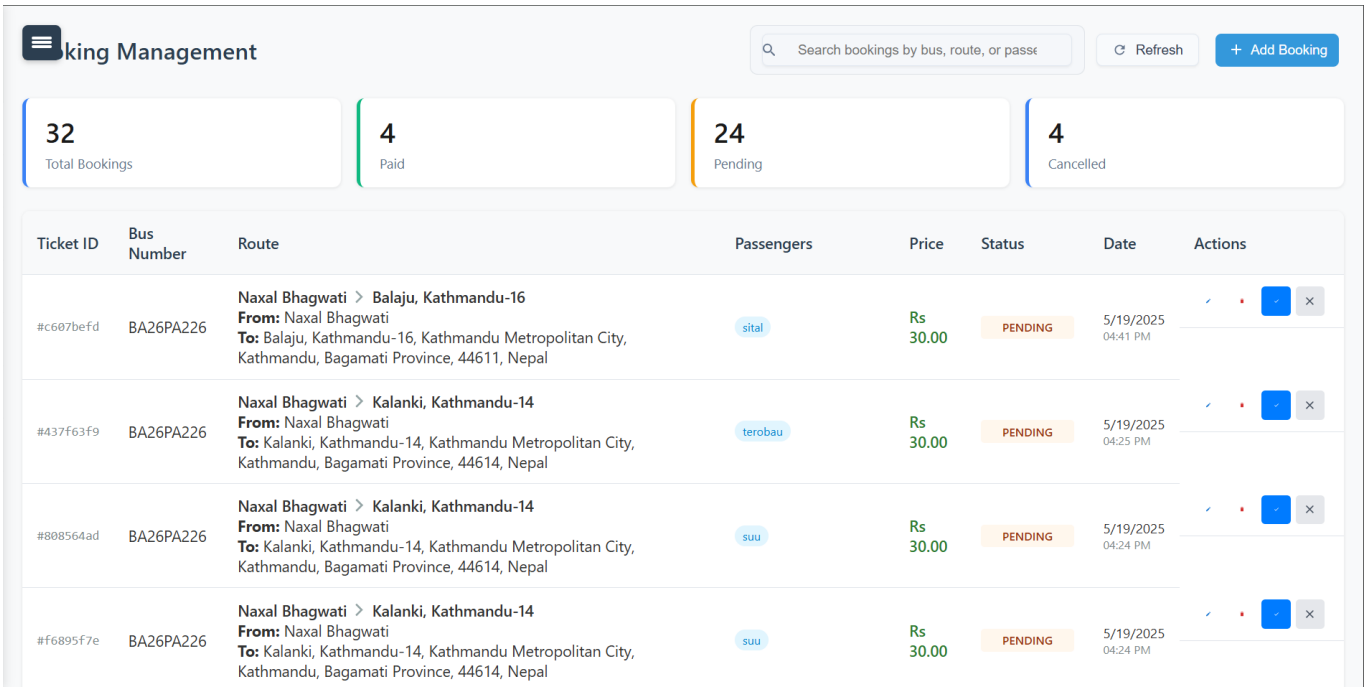


Figure 37: Booking management for admin

5.2.15 Payment using Khalti

Test Case	TC_08
Objective	Payment Integration using Khalti
Action	Pay via Khalti after booking
Expected Results	After booking the payment should and the status should be displayed as paid
Actual Results	Successfully paid and booked the ticket
Conclusion	Success

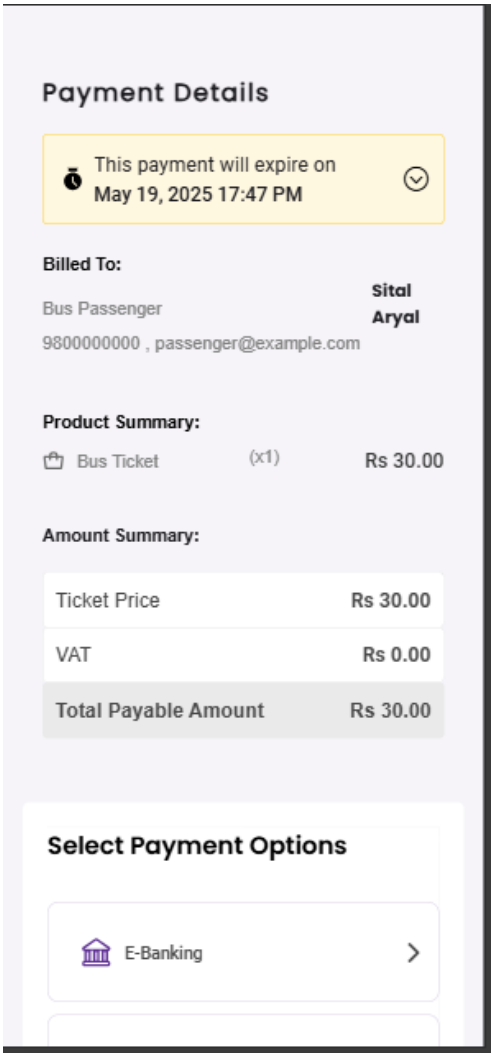


Figure 38: Khalti payment UI

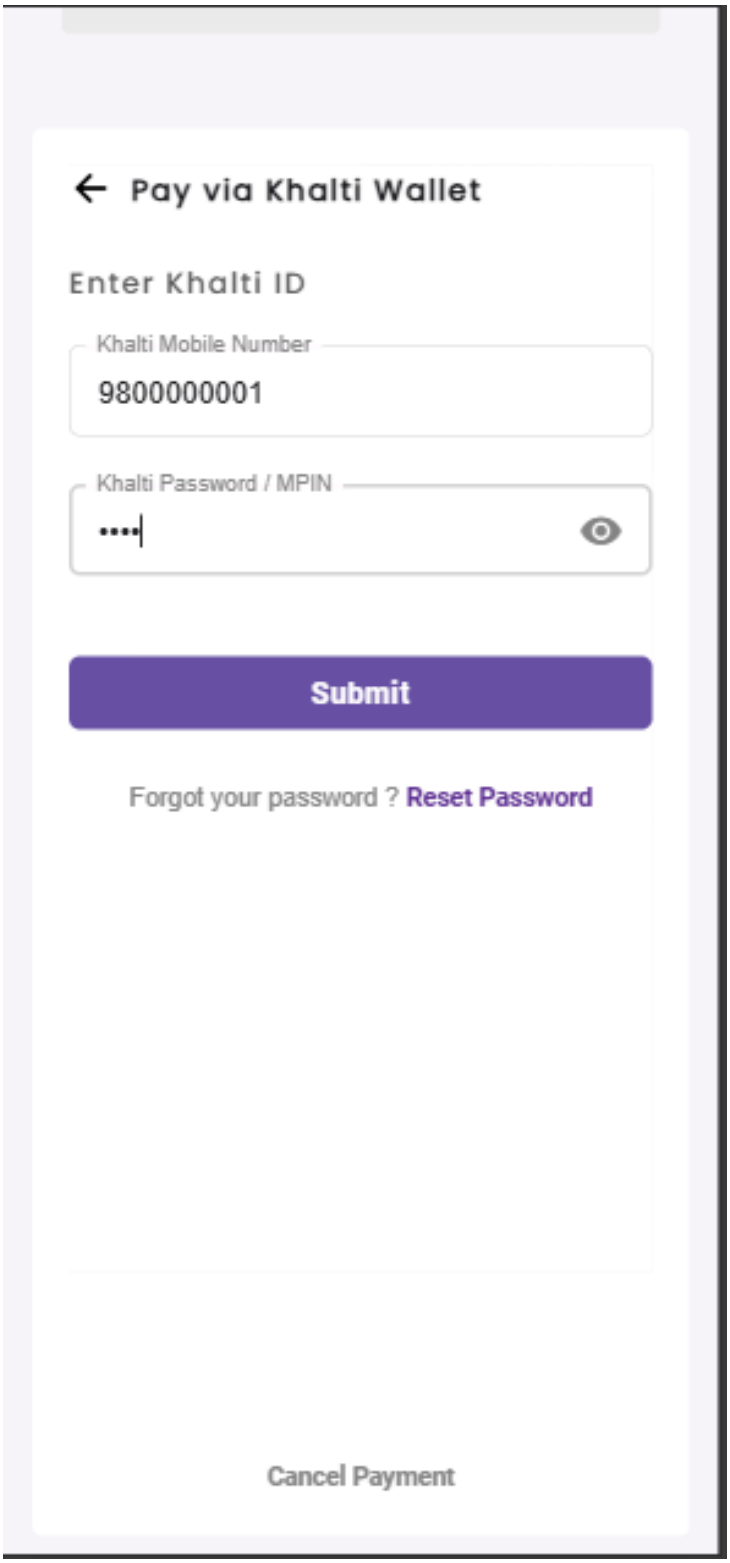


Figure 39: Valid credentials for payment

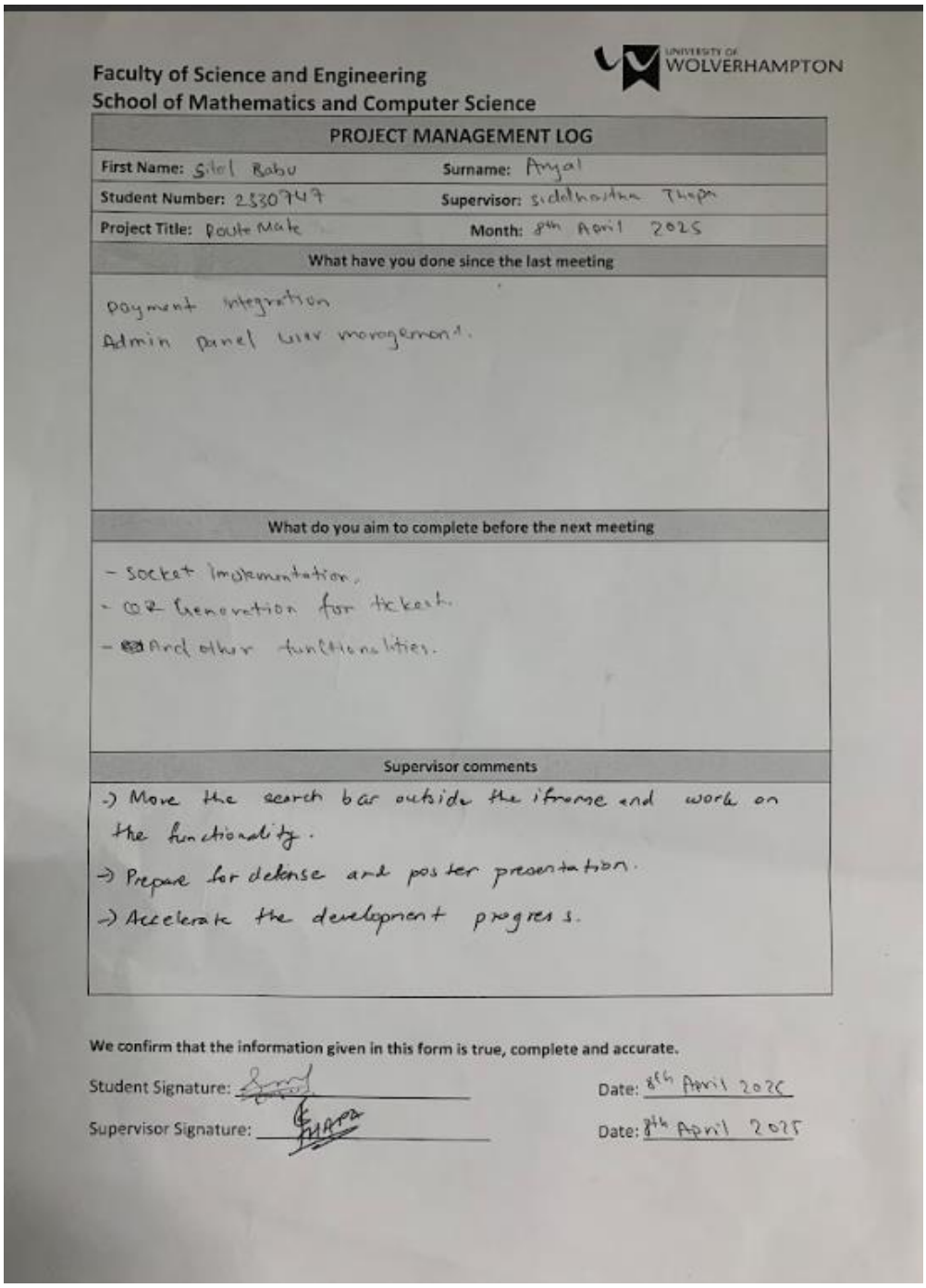
5.2.16 Rating and Review System

Test Case	TC_08
Objective	To give rating after the successful.
Action	Successfully rate the driver
Expected Results	Users can rate the driver and edit it
Actual Results	Successfully paid and booked the ticket
Conclusion	Success

6. Evidence of Project Management

6.1 Log sheet

These log sheets are the track of the project work completion since the last meeting, attended meeting and works needs to be done before next meeting and supervisor remarks. All the log sheets are attached below.



Faculty of Science and Engineering
School of Mathematics and Computer Science

UNIVERSITY OF WOLVERHAMPTON

PROJECT MANAGEMENT LOG

First Name: Silol Babu	Surname: Anyal
Student Number: 2330747	Supervisor: Siddhartha Thapa
Project Title: Route Mate	Month: 8th April 2025

What have you done since the last meeting

Payment integration
Admin panel user management.

What do you aim to complete before the next meeting

- Socket Implementation.
- QR Generation for tickets.
- And other functionalities.

Supervisor comments

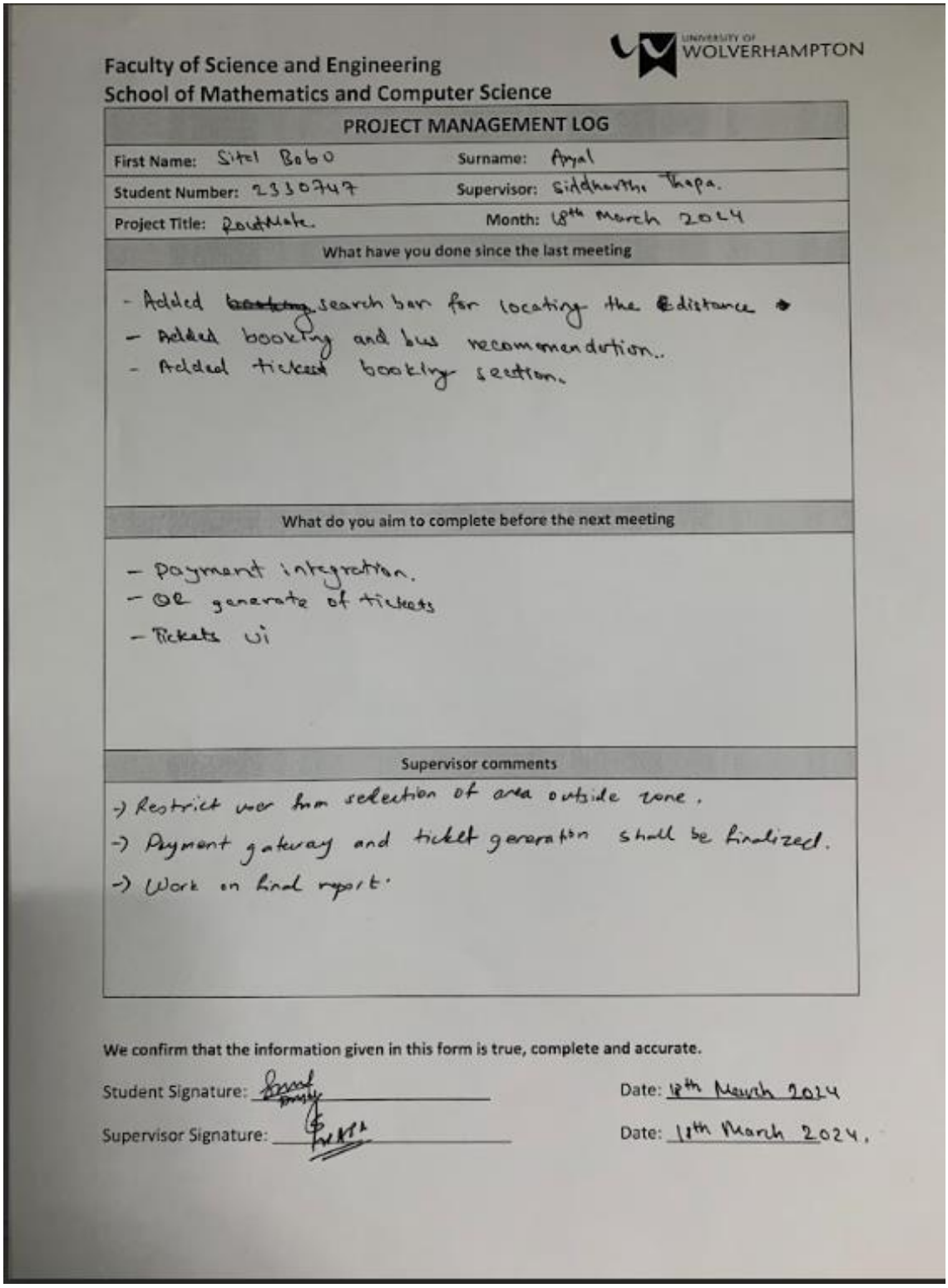
-> Move the search bar outside the iframe and work on the functionality.
-> Prepare for defense and poster presentation.
-> Accelerate the development progress.

We confirm that the information given in this form is true, complete and accurate.

Student Signature: _____ Date: 8th April 2025

Supervisor Signature: _____ Date: 8th April 2025

Figure 40: Log sheet 1



Faculty of Science and Engineering
School of Mathematics and Computer Science

UNIVERSITY OF WOLVERHAMPTON

PROJECT MANAGEMENT LOG

First Name: Sitel Bobo	Surname: Ayal
Student Number: 2330747	Supervisor: Siddharth Thapa.
Project Title: RoutMate.	Month: 18th March 2024

What have you done since the last meeting

- Added ~~booking~~ search bar for locating the distance
- Added booking and bus recommendation..
- Added ticket booking section.

What do you aim to complete before the next meeting

- Payment integration.
- QR generate of tickets
- Tickets UI

Supervisor comments

-) Restrict user from selection of area outside zone.
-) Payment gateway and ticket generation shall be finalized.
-) Work on final report.

We confirm that the information given in this form is true, complete and accurate.

Student Signature: _____ Date: 18th March 2024

Supervisor Signature: _____ Date: 18th March 2024.

Figure 41: log sheet 2

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School of Mathematics and Computer Science

UNIVERSITY OF WOLVERHAMPTON

PROJECT MANAGEMENT LOG

First Name: Sital Babu	Surname: Aryal
Student Number: 2330747	Supervisor: Siddhartha Thapa.
Project Title: Routemate.	Month: 11th March 2025

What have you done since the last meeting

- Implamented socket.io to display real time location
- Improved ui
- location displayed in map.

What do you aim to complete before the next meeting

- Enable driver panel
- Implement socket.io to display realtime data and store it.
- display driver location if possible.

Supervisor comments

→ Socket needs to be implemented on frontend side and development progress needs to be accelerated.
→ Booking feature need to be implemented first.
→ Final draft need to be compiled.

We confirm that the information given in this form is true, complete and accurate.

Student Signature: _____ Date: 2025 march 11th.

Supervisor Signature: _____ Date: 2025 march 11th.

Figure 42: log sheet 3

Faculty of Science and Engineering
School of Mathematics and Computer Science

WOLVERHAMPTON

PROJECT MANAGEMENT LOG

First Name: Jital Babu	Surname: Ayal
Student Number: 2330747	Supervisor: Siddha Ahu Thapa.
Project Title: Routemate	Month: 4th March 2025

What have you done since the last meeting

- I added new fields to signup
- Professionalism report.
- Map integration
- Admin panel ui.

What do you aim to complete before the next meeting

- Register user displays in Admin Panel
- Driven ui enhancement
- use real time location data for tracking live location

Supervisor comments

-> Development progress need to be increased overally, looks good.
-> Professionalism report needs to be updated as recommended.

We confirm that the information given in this form is true, complete and accurate.

Student Signature: _____ Date: 4th March 2024

Supervisor Signature: _____ Date: 4th March 2024

Figure 43: log sheet 4

Faculty of Science and Engineering
School of Mathematics and Computer Science

UNIVERSITY OF WOLVERHAMPTON

PROJECT MANAGEMENT LOG

First Name: Sital Babu	Surname: Anyal
Student Number: 2330747	Supervisor: Siddhartha Thapa
Project Title: Route work (Bus track tracking system)	Month: 10th Dec 2024

What have you done since the last meeting

I've created UI/UX design for passenger panel which includes login sign up, verification, Home page, booking details and profile overview. I also set up project.

What do you aim to complete before the next meeting

I aim to complete frontend of login page and make the login page responsive for all the devices.

Supervisor comments

- start login signup UI and also complete login authentication if possible.

We confirm that the information given in this form is true, complete and accurate.

Student Signature: _____ Date: 10th Dec 2024

Supervisor Signature: _____ Date: 10th Dec 2024

Figure 44: log sheet 5

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School of Mathematics and Computer Science

UNIVERSITY OF WOLVERHAMPTON

PROJECT MANAGEMENT LOG

First Name: Sital Babu Surname: Aryal

Student Number: 2330747 Supervisor: Siddhartha Thapa

Project Title: Rukmek (A bus Route tracking system) Month: 28th Jan 2025

What have you done since the last meeting

- made Forgotpassword page
- Added document upload page for bus driver
- Improved forgot passwon ui
- Role switch button navigated to document verification page.

What do you aim to complete before the next meeting

- All the document verification page
- Admin panel gets start
- Admin will be able to verify the documents after reviewing.

Supervisor comments

→ Development progress needs to improved based on suggestion.
→ Architect design needs to be done.

We confirm that the information given in this form is true, complete and accurate.

Student Signature: _____ Date: 28th Jan 2025

Supervisor Signature: _____ Date: 28th Jan 2025

Figure 45: log sheet 6

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School of Mathematics and Computer Science

UNIVERSITY OF WOLVERHAMPTON

PROJECT MANAGEMENT LOG

First Name: Sital Babu Surname: Aryal

Student Number: 2330747 Supervisor: Siddhartha Thapa.

Project Title: RoutMak(Bus Route tracking system) Month:

What have you done since the last meeting

- Added bus document ~~page~~ upload page
- Made the camera working while uploading the picture.
- ~~to~~ ~~phone~~ App access ~~to~~ the ~~storage~~ file to upload the picture.
- Admin panel UI

What do you aim to complete before the next meeting

- Map integration for users ~~and~~ and drivers.
- ~~Fu~~ functional admin panel.
- documents get stored in database.

Supervisor comments

-> Map integration need to be worked upon.
-> Driver registration form needs to be functional.
-> Start working on professionalism report.

We confirm that the information given in this form is true, complete and accurate.

Student Signature: _____ Date: 2025 feb 11

Supervisor Signature: _____ Date: 2025 feb 11

Figure 46: log sheet 7

Faculty of Science and Engineering
School of Mathematics and Computer Science

UNIVERSITY OF WOLVERHAMPTON

PROJECT MANAGEMENT LOG

First Name: Sital Bobu	Surname: Anyal
Student Number: 2130744	Supervisor: Siddhartha Thapa
Project Title: RouteMate (Lux Route Tracking System)	Month: 5th Jan 2025

What have you done since the last meeting

- Improved ~~a~~ CSS of login and signup page
- Added a home page after login:
- Added hamburger menu and ALLO Role switch button.

What do you aim to complete before the next meeting

- ~~Sooo~~ Syne the login info in profile page
- Driven verification and ~~route doesn't~~ role assignment after verification.

Supervisor comments

→ Login/signup implementation looks goods.
→ UI improvement looks good.

We confirm that the information given in this form is true, complete and accurate.

Student Signature: _____ Date: 5th Jan 2025

Supervisor Signature: _____ Date: 5th Jan 2025

Figure 47: log sheet 8

Faculty of Science and Engineering
School of Mathematics and Computer Science

UNIVERSITY OF WOLVERHAMPTON

PROJECT MANAGEMENT LOG

First Name: Sital Babu	Surname: Pryol
Student Number: 2330747	Supervisor: Siddartha Thapa.
Project Title: Routemate (Bus Route tracking system)	Month: 22th Dec 2024

What have you done since the last meeting

I've created login page using ~~next~~ ~~next~~ react native expo and also create ui according to my figma design.

What do you aim to complete before the next meeting

I aim to complete proper login signup authentication for my app and also create home page frontend if possible.

Supervisor comments

→ login / Signup needs to be implemented with database connection.

→ Literature review needs to be started.

We confirm that the information given in this form is true, complete and accurate.

Student Signature: _____ Date: 22th Dec 2024

Supervisor Signature: _____ Date: 22th Dec 2024.

Figure 48: log sheet 9

Faculty of Science and Engineering
School of Mathematics and Computer Science

UNIVERSITY OF WOLVERHAMPTON

PROJECT MANAGEMENT LOG

First Name: Sital Babu	Surname: Aryal
Student Number: 2330747	Supervisor: Siddhartha Thapa
Project Title: Routemate (Bus route tracking system)	Month: 29th December.

What have you done since the last meeting

- I made login signup authentication. using node and postres for database.

What do you aim to complete before the next meeting

- complete the UI of homepage and assign roles too.

Supervisor comments

- Literature review needs to be worked upon.
- Push progress to git.
- work on role based authentication.

We confirm that the information given in this form is true, complete and accurate.

Student Signature: _____ Date: 29th Dec 2024

Supervisor Signature: _____ Date: 29th Dec 2024.

Figure 49: log sheet 10

6.2 Gantt chart

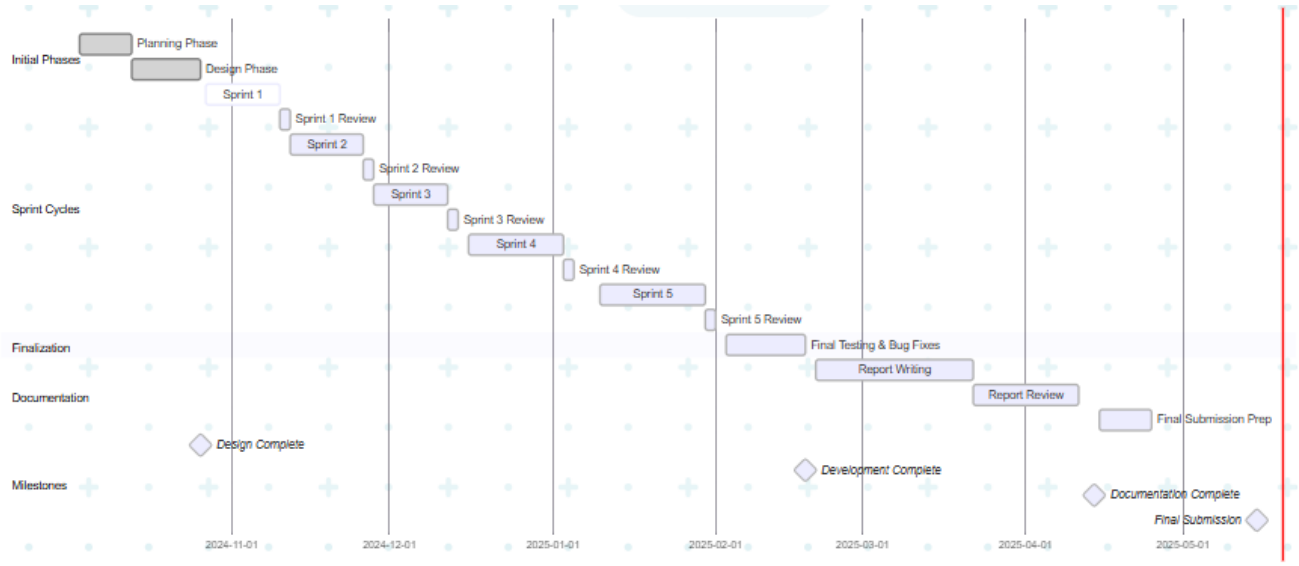

Figure 50: Gantt chart

7. Conclusion

The RouteMate Bus Route Tracking System has achieved all project targets including academic questions which implies mobile-based solutions can improve public transit efficiency and reliability in metropolitan environments like Kathmandu.

Key Achievements

- RouteMate delivers a friendly user system through which passengers can follow buses in real time and find nearby stops and chart routes efficiently to meet their requirements for enhanced accessibility and reliability.

- Real-Time Tracking benefits from mobile network signals and Wi-Fi in order to provide precise bus location tracking and estimated arrival times within GPS-limited boundaries which matches the target to address public transport tracking problems.

- Through digital integration the system offers features that combine digital ticketing with contactless payment systems from eSewa and Khalti in addition to emergency alerts to achieve modern transport operations.

- The implemented project demonstrated how mobile technology supplied solutions to bridge operator-passenger communication gaps and reduce waiting time uncertainties while motivating passengers to select public transport instead of private vehicles. The research results directly addressed the academic inquiry.

Discoveries and Lesson Learned

- The system demonstrated that real-time tracking procedures do not need to depend exclusively on expensive GPS technology because alternative network data sources can be used effectively.

- The systematic enhancements and user education about digital payment adoption remain crucial because of the clear benefits together with internet dependency and initial resistance from users.

- The iterative approach of Agile Methodology enabled team members to acquire ongoing user feedback which resulted in adapting the development until achieving effective solutions.

A well-designed mobile application through RouteMate has successfully achieved all three goals: enhanced public transportation efficiency and user satisfaction and improved operational transparency. The system addresses typical user issues about inconsistent schedules and insufficient real-time information which helps minimize traffic congestion and environmental hazards. Future work should let people connect through additional networks in addition to improving data precision by creating stronger partnerships with local transit agencies to increase adoption.

This study demonstrates how technology affects urban transportation because it presents adaptable solutions for cities dealing with comparable transportation issues. The achievement of RouteMate delivers a reliable base to propel advanced development of intelligent public transportation solutions.

8. Critical Evaluation of the Project

RouteMate has achieved its objectives by introducing a real-time bus and ticket tracking system. All you need is the mobile app to do everything from tracking your fleet to handling payments and booking. When you use mobile networks and Wi-Fi in addition to GPS, it becomes a reliable option for locations where resources are low. Since the system has roles for passengers, drivers and admins, it can serve all interested individuals. All in all, the report reveals the benefits mobile solutions have for users of public transport.

Yet, we met a few obstacles while working on the project. This method of communication requires a good internet connection that isn’t always found everywhere. Additionally, some transport operators may take time before deciding to use the system. Using digital ticketing and touch-free payment may take some time for a few users to get used to. Tracking may not be accurate when there are technical issues, server problems or a weak GPS signal. However such problems can be handled if we provide support, education and updates and they do not prevent the project from finding success.

8.1 System Evaluation

The system was built using modern tools like React Native, Node.js, and PostgreSQL, making it responsive and cross-platform compatible. Testing was done for major functions including login, tracking, booking, and payment, and all tests passed successfully. The system uses Leaflet and OpenStreetMap for map integration, which displayed real-time bus locations effectively. Users receive updates on bus locations, booking confirmations, and payment statuses. The interface is simple and clean, suitable for all types of users, even those not very familiar with technology.

Still, there is always more that can be done to make things better. So, if the network is slow, updates on the map may be delayed and updates on drivers' locations rely on the drivers having a connection online. Furthermore, future updates may add route optimization and voice assistance to the GPS software. While the security of personal information and transactions is adequate, new updates may be necessary to maintain its reliability. If some adjustments and routine care are given, the system will likely become both better and more widespread.

8.2 Self-Reflection

I have gained a lot of insights while using RouteMate. I got better at using React Native and developing applications with Node.js and PostgreSQL on the backend. I discovered that agile methodology helps break a big project into easy-to-manage parts. I was able to make fewer errors and quickly fix them because I met with my supervisor on a regular basis. When I designed wireframes and built real-time options, I gained trust in coming up with unique solutions for difficult issues.

I learned to work on soft skills such as communication, time management and solving problems. Handling jobs that require technology and those that affect users taught me good user experience is tied to good functionality. I discovered that each sprint relied on testing, feedback and improvements. I learned that having small features like maps and notifications on my phone is useful day to day. On the whole, I am proud of where I am and eager to upgrade and improve things even more.

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